Smart Camp Box

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The immediate aftermath of natural disasters frequently disrupts critical social infrastructure, leaving millions of children in states of neurological suppression that hinder formal learning. Trauma-induced hyperactivation of the amygdala suppresses prefrontal cortex functions essential for academic retention, rendering conventional educational delivery ineffective and potentially counterproductive for displaced children [1]. Cyclone Ditwah (November 2025, Sri Lanka) illustrates this crisis: 890,000 children were affected, 1,682 schools were damaged, and teacher-to-student ratios reached 1:77, with no AI-driven educational systems deployed in the emergency response [2]. Existing educational technologies assume learners are psychologically prepared to engage with content; no system currently assesses combined psychological and attentional readiness before instruction or monitors individual children for persistent welfare distress [3],[4].

This study introduces the Psycho-Attentional Gated Educational System (PAGES), an offline-first IoT web application designed to assess readiness among Grade 3-4 children (ages 7-9). PAGES employs a three-source behavioral fusion model integrating a gamified inhibitory control task, passive interaction signals, and a rolling historical baseline. A binary gate decision determines access to standardized curriculum content per session. A two-phase anomaly detection pipeline applies Z-score statistical thresholding followed by per-child Isolation Forest modeling to monitor behavioral patterns and generate plain-language welfare alerts for admin.

The system is implemented using a MERN stack extended with Python and scikit-learn, SQLite for offline edge storage, and deployed on Raspberry Pi 5 hardware without internet dependency. A pilot study with 20–30 students from Cyclone Ditwah-affected schools under the Homagama Zonal Education Division is expected to demonstrate significant assessment sensitivity ($p < 0.05$), gate accuracy above 80%, curriculum alignment exceeding 90%, welfare alert precision of 0.75, and recall of 0.70, and complete offline functionality with full cloud synchronization fidelity. Future work will evaluate PAGES in active emergency deployments, extend anomaly detection with clinical validation, and integrate complementary Smart-Camp Box components.

Keywords: *post-disaster education, psycho-attentional gating, anomaly detection, Isolation Forest, offline IoT, child welfare monitoring*

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Acknowledgment is also extended to the Smart-Camp Box research group, through which PAGES operates, as well as appreciation to all members of the PAGES team, whose parallel academic efforts have been incorporated into the system architecture.

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LIST OF ABBREVIATIONS

posttraumatic stress disorder - PTSD

Psycho-Attentional Gated Educational System - PAGES

National Institute of Education - NIE

United Nations Educational, Scientific and Cultural Organization - UNESCO

United Nations Children's Fund - UNICEF

National Disaster Management Centre - NDMC

Zonal Education Office - ZEO

Zone of Proximal Development - ZPD

Continuous Performance Test - CPT

Organisation for Economic Co-operation and Development - OECD

Information Technology Services - ITS

United Nations - UN

Artificial Intelligence - AI

Internet of Things - IoT

Design Science Research - DSR

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1. INTRODUCTION

1.1 Background and Context

1.1.1 Problem Definition and Contextual Framing

The intersection of large-scale natural disasters forced displacement, and psychological trauma among children represents one of the most complex challenges that remains insufficiently addressed in contemporary humanitarian technology design. When Cyclone Ditwah made landfall on Sri Lanka's eastern coast on November 28, 2025, it produced not merely a physical emergency but a cascading educational and psychological crisis affecting an estimated 890,000 children across nine provinces [1]. In a period of just 72 hours, 1,682 primary and secondary schools were damaged or rendered structurally unsafe, displacing thousands of children into temporary relief camps. In these settings, the foundational conditions for learning, psychological safety, cognitive availability, and access to supportive learning structures were not only absent. Still, they were a predictable outcome of mass migration due to natural disasters [2].

The challenge of post-disaster education is not purely logistical. It is deeply psychological and attentional in nature. Children affected by disaster exhibit clinically significant levels of post-traumatic stress, anxiety, grief, and emotional dysregulation [3]. These psychological states directly impair cognitive functioning, attentional capacity, and learning readiness, meaning that even when educational resources are physically restored, a traumatized child may be present in a learning environment yet remain neurologically and psychologically unavailable to learn [4]. Neuroscientific evidence confirms that chronic stress responses in young children suppress prefrontal cortex activity, the brain region responsible for attention, working memory, and learning, making cognitive engagement physiologically impossible under conditions of unresolved psychological distress [5].

Many children exposed to traumatic events develop distress symptoms, which can represent normal reactions to abnormal circumstances [6]. These reactions may include fears of recurrence, anxiety, sadness, grief, sleep disturbance, difficulty concentrating, anger, behavioral outbursts, or developmental regression [7],[8]. Increases in risk-taking behaviors, such as alcohol and substance use, have also been described among adolescents after disasters [9]. Some children will experience symptoms that meet criteria for psychiatric disorders, such as acute stress disorder and posttraumatic stress disorder (PTSD).

Evidence is built on the development of posttraumatic stress symptom profiles and trajectories in children exposed to disasters. A study of eight studies on posttraumatic stress symptom profiles in children exposed to natural disasters found that between 4% and 23% had a chronic posttraumatic stress symptom profile, including elevated levels of PTSD over time. An additional 7% to 27% had a recovery profile, characterized by an initial elevation of posttraumatic stress symptoms followed by decreasing levels over time. A significant percentage, 45% to 79%, had a resilience profile characterized by low levels of posttraumatic stress symptoms over time [10].

The problem addressed by this component is not reducible to a shortage of educational content or a lack of qualified teachers and volunteers, though both are documented realities of the Ditwah response. The deeper problem is one of system design. Current humanitarian education responses are architecturally misaligned with the neurobiological reality of how trauma affects the learning ready brain. Existing digital education tools deployed in disaster and refugee settings, including Kolibri (Learning Equality), KA Lite, and Remote Area Community Hotspot for Education and Learning (RACHEL), operate on the implicit assumption that the learner is psychologically prepared to receive and process academic content. This assumption, which may be reasonable in stable educational contexts, is neurologically unfounded for acutely traumatized children [11]. Research in trauma neuroscience has established with considerable consistency that sustained activation of the amygdala's threat-detection circuitry during post-traumatic stress suppresses activity in the prefrontal cortex [12],

the neural region responsible for working memory, executive function, and the higher-order cognitive processes that academic learning requires. In practical terms, a child in acute post-disaster distress cannot learn mathematics or language, not because the content is inappropriate or the teacher inadequate, but because the neural architecture that would process and retain that content is functionally offline.

Psycho-Attentional Gated Educational System (PAGES) is designed to address this misalignment directly. Rather than providing educational information to all children regardless of their psychological and attentional states, PAGES uses a psycho-attentional gating architecture wherein the combined psychological and attentional states of the child, as measured by passive behavioral interaction signals, determine whether the child is allowed to engage with cognitive educational information at all. The psychological and attentional states of the child are not an additional process that occurs concurrently with educational engagement; rather, they are a measurable process that must be satisfied before educational engagement is allowed, which is supported by neuroscientific studies on the effect of trauma on cognition [12]. It is noteworthy that PAGES does not implement adaptive learning, wherein educational content is modified according to the performance of the child. Rather than this approach, PAGES uses a gating architecture wherein readiness is measured, a threshold is evaluated, and only upon successful completion of this process is the child allowed to engage with standardized cognitive educational content.

1.1.2 Affected Populations, Magnitude of Impact, and Stakeholder Analysis

A. Primary Target Population: Children in Primary Education (Ages 7-9)

The main population of interest for the PAGES component is children aged 7 to 9 years, which equates to Grades 3 and 4 of primary education in Sri Lanka, as run by the Ministry of Education and National Institute of Education (NIE). This age range is more than just a statistical construct; it is supported by two lines of evidence that converge to provide a strong rationale for its selection.

Merzenich, along with other recent studies in the field of developmental neuroscience, has identified that the range of 6 to 11 years old represents the peak period of synaptic density and the formation of neural paths in the human brain [13]. From a cognitive developmental perspective, Piaget's theory of cognitive development places children aged 7-11 in the Concrete Operational Stage of the period during which logical thinking, categorization, cause-and-effect reasoning, and structured learning first become neurologically viable [14]. Grade 3 specifically represents the internationally recognized 'Grade 3 Gateway,' the critical checkpoint by which foundational literacy and numeracy must be established if long-term learning outcomes are to be protected [15]. United Nations Educational, Scientific and Cultural Organization (UNESCO) and the World Bank both identify Grades 3 and 4 as the single most consequential intervention point in primary education, noting that children who have not achieved foundational literacy by this stage face a compounding deficit that widens progressively through secondary school and into adult economic outcomes [16][17]. Grade 4 extends the window through the completion of the Concrete Operational Stage's most productive phase, enabling the system to capture children at both the entry point and the consolidation point of this critical developmental window.

From a psychological readiness perspective, Blair & Razza's seminal work clearly indicates that self-regulation and attentional readiness are biological pre-requisites for learning among children aged nine or less, not merely facilitators of learning [18]. This is particularly pertinent to the context of the proposed PAGES deployment scenario, a zero-internet, volunteer-staffed scenario following a disaster situation, because, unlike older children who can maintain their own interest through intrinsic academic motivation, Grades 3 and 4 children require psychological readiness before any potentially beneficial

cognitive content is presented to them. The gating model is not only theoretically justified but essential for this age cohort.

B. Magnitude of Impact: Systemic Failures Beyond the Classroom

While children constitute the primary end users of PAGES' educational and psychological support functions, an accurate account of the problem's magnitude requires examination of the broader systemic failures that render conventional educational responses structurally inadequate in the disaster camp context. These failures, documented in both the Ditwah field reports and the academic literature, collectively establish that technological mediation is not a supplementary convenience but a structural necessity.

The Ditwah response indicates that the teacher-to-student ratio is approximately 1:77, based on 308 teachers serving 23,695 students in internal displacement camps [19]. This teacher-to-student ratio makes it operationally infeasible to employ methods that require teacher observation, assessment, and/or differentiation of students. A teacher cannot simultaneously attend to 77 students to ascertain each student's cognitive state, detect signs of psychological distress, and adjust the presentation of the subject matter to the needs of individual students. Accordingly, the gating algorithmic layer provided by the PAGES is not an improvement over the status and it is a replacement for human ability that cannot exist in the context in which it is being used.

The problem is further compounded by the trauma profile of the teaching staff. UNICEF Situation Report No. 8 stated that the total number of teachers and school principals in the affected provinces identified as trauma-affected individuals in need of psychosocial support was estimated at 28,900 [2]. This problem presents an ethical dilemma because human actors in the children's process of healing and moving forward are the same ones who need the same process of healing. It is not only optimistic but impossible for trauma-affected teachers to provide trauma-informed teaching to 77 students without technological support.

The wider global context adds further weight to the significance of this locally focused research. According to UNESCO, over 222 million children in 93 countries require humanitarian support for their educational needs each year, and the vast majority of those are in low and middle-income countries that have limited digital connectivity [20]. The systematic review carried out by Crompton et al. [21], which reviewed 60 studies in 48 countries, explicitly highlighted the need for research in this area as a priority gap in the evidence base. They highlighted the fact that internet-based solutions dominate the evidence base, whereas solutions for low-connectivity environments are vastly underrepresented. The issue of a lack of learning tools that can function in an offline capacity is therefore a priority research need.

C. Stakeholder Consultation: Methodology and Institutional Context

The results of the consultations are reflected in the design rationale of the component in the following table: Table 1.

Table 1: Stakeholders and users consultation summary

Stakeholder	**Role / Position**	**Key Pain Points Identified and observations**	**Implications for PAGES**
Mr. P.S. Priyanath Perera	Divisional Secretary, Kolonnawa (Administrative authority)	• Data Gathering and Entering • Immediate post-disaster education is infeasible due to psychological shock	• Validates PAGES's requirement to register and track individual children from camp arrival

		• Technical human resource limitations constrain digital deployment • Most of the students stopped their education after grades 8 or 10. Suggest collaborating with them on our project to motivate them. • Some children lack support from their families for education. No better motivation for their life and education. • Children are not in a comfortable home, and they may feel unsafe. Fathers may be at home to protect their properties, and they may feel that they are unsafe without their fathers.	and provide physical support when needed. • Giving hope to children that education is not disrupted due to disaster and giving psychological support when required.
Mr. Dhananjaya	Senior Field Officer, National Disaster Relief Services Centre (NDRSC)	• Inaccurate camp data resource misallocation due to non-camp residents accessing supplies. • Camp duration can be more than 2-4 weeks in the Badulla district, and education support is more required.	• Directly validates PAGES's phase-gated design: psychological stabilization before academic content. • Informs offline-first, low-technical-barrier deployment requirement. • Giving hope to children that education is not disrupted due to disaster and giving psychological support when required.
Mrs. K. M. A.N. Bandara	Assistant Zonal Educational Director (School Administration), Homagama	• Research permission granted; school access to Grade 3-4 students confirmed for pilot study	• Pilot study access secured for Design and Validate a Combined Psycho-Attentional Assessment Layer and Develop and

			Evaluate a Machine Learning Anomaly Detection System for Child Welfare Monitoring evaluation; ethical clearance pathway confirmed
Ms. Umanjalee Amunumulla	Primary Teacher, Sri Rahula Maha Vidyalaya, Mulleriyawa (Community stakeholder-education)	• Schools were closed for more than 2 weeks. • Some students showed poor attendance at school for a period. • Personal trauma affecting facilitation capacity is limited, and no structured guidance system is available.	• Validate the volunteer facilitator dashboard • Integrate child-friendly digital tools for resilience • Education with limited resources. • Consider the psychological level before giving education.
Disaster-Affected Children (Ages 7-9, Grades 3 and 4)	Student	• Psychological shock prevents immediate learning engagement. • Unfamiliar camp environment disrupts routine and sense of safety • Hopeless for the future and felt fear	• Confirms phase-gated design necessity. • Educational and psychological support. • Give physical support if recognized by the system. • Help to improve the cognitive level of each.

1.1.3 Statistical Evidence Base

The following quantitative data, verified through primary humanitarian sources documenting the Cyclone Ditwah response, establishes the empirical basis for the research problem:

- 890,000+ children affected across nine provinces in Sri Lanka (UNICEF Sri Lanka Humanitarian Situation Report No. 8, February 2026) [1]
- The total number of primary and secondary schools structurally damaged or unsafe in the affected area was 1,682 (UNICEF Situation Report No. 7, January 2026) [2]
- Between 15-20% of the affected schools sustained structural damage precluding immediate educational use (UN Sri Lanka SitRep #7, 2026) [2]
- 500 schools repurposed as displacement shelters, requiring complete evacuation, cleaning, and physical repair before educational function could be restored (Sri Lanka Red Cross Situation Report No. 8, November 2025) [22]
- 308 teachers serving 23,695 students in documented IDP camp configurations, representing a teacher-student ratio of approximately 1:77 [19]

- 28,900 teachers and school principals identified as trauma-affected and requiring psychosocial support, the very population tasked with delivering children's educational recovery [2]
- Only 36 operational child-friendly spaces served 1,600 children out of a total of almost 900,000 people in need of such facilities, indicating a coverage rate of less than 0.2%. (UNICEF, 2026) [1]
- The total number of operational child-friendly spaces was 45, which served 28,504 children and 11,377 caregivers out of the total number of children in need of such facilities. The coverage rate was less than 4%. (UN SitRep #7, 2026) [2]
- The number of children with special needs who required specially adapted learning material was 900. There was no digital adaptive solution deployed at any point in the response (UNICEF, 2026) [1]
- There was no Artificial Intelligence (AI) or adaptive learning technology deployed in response to Cyclone Ditwah. This is the primary technology used to address research problems. [1][2][23]

The final statistics above are particularly significant as research justification. Despite significant levels of humanitarian funding, the overall educational response to the crisis involving 890,000 children was channeled solely through physical kits, human resources, and static, non-adaptable digital media. This digital deficit, however, should not be regarded as an isolation, given the overall pattern that has been identified in multiple disaster scenarios wherein the creation of intelligent, offline-capable, psychologically aware educational systems has lagged far behind the established humanitarian need.

1.1.4 Critical Analysis of Existing Solutions and Their Limitations

A rigorous assessment of the Cyclone Ditwah educational response reveals that existing humanitarian education interventions, while representing genuine and well-resourced efforts, share a set of structural limitations that collectively constitute the justification for the proposed technological approach.

Table 2: Existing Solutions and Their Limitations

Existing Solution	**What It Provides**	**Structural Limitation**
UNICEF Learning Kits (81,300 distributed post-Ditwah) [1]	• Physical education Materials • Age-grouped content • Individualized pacing	• Non-digital • No psychological state detection • Fixed content regardless of trauma-induced cognitive regression • No individual monitoring
Child-Friendly Spaces -45 CFS serving 28,504 children [2]	• Safe play-based learning • Group psychosocial support	• Dependent on trained human facilitators • Covers <4% of affected children • No individual digital data collection • No welfare alert mechanism
Kolibri (Learning Equality)-deployed in	• Offline Raspberry Pi content delivery	• No psychological or attentional pre-assessment

Jordan refugee camps [24]	• Progress tracking • Curriculum support	• No gating model • No welfare anomaly detection • Static content navigation is not algorithmically personalized
Save the Children Digital Packs [25]	• USB/SD card content for CFS; interactive stories; mindfulness games for group use	• Static and non-adaptive • no individual trajectory tracking
TeamUp Methodology- Save the Children / CCH, Sri Lanka [21]	• Structured play and movement for psychosocial recovery; evidence-based for PTSD reduction	• Discrete psychosocial program entirely separates from educational content • No information about the psychological state is shared with the educational context • Structural fragmentation
WHO mhGAP Offline Application [25]	• Mobile tools for healthcare providers • Child assessment modules • Coping strategies	• Designed for adult professional use, not for direct child interaction • No educational function • Clinical rather than educational-recovery focus
Traditional Adaptive Learning Systems (DreamBox, Knewton) [21]	• Content difficulty adjustment based on academic performance data	• Assumes child is cognitively ready; does not assess psychological or attentional state before learning begins • Not designed for disaster contexts • Internet-dependent
Psychological First Aid Programs- 621 workers trained post-Ditwah [2]	• Psychological stabilization reduces PTSD and anxiety via trained volunteers	• Requires trained human facilitators • Not integrated with educational systems • Cannot continuously monitor individual children across sessions • Not scalable to 890,000 children

1.1.5 Proposed Solution Overview

PAGES makes five integrated original contributions:

1. A Combined Psycho-Attentional Assessment Layer is proposed to quantify the psycho-attentional readiness of each child using three indirect behavioral measures, thus avoiding direct self-reporting by the child. Specifically, the psycho-attentional readiness score is composed of the following components:

a) A gamified Go/No-Go paradigm task [26] (50% weight): In this task, the child responds to the target animal by tapping on the screen, avoiding responses to the no-go stimuli, over a 90-second interval. The accuracy, false tap rate, reaction time, and reaction time variance of the responses are recorded.
b) Passive behavioral interaction signals [27] (30% weight): These comprise the idle time ratio, rapid click count, time to first interaction, and session completion ratio.
c) A pre-computed rolling baseline score stored within the local profile of the child, which is updated using a weighted rolling average after each session (New Baseline = Old Baseline × 0.8 + Today's Score × 0.2) (20% weight).

2. A Psycho-Attentional Gating Model that determines, based on the composite score against a defined threshold, whether a child is in a neurologically viable state to engage with cognitive learning content. Children below threshold receive a brief calming or re-engagement support activity and are re-assessed before any cognitive content is offered. Children above the threshold proceed directly to the cognitive layer. This gating architecture is grounded in Bloom's Mastery Learning Model [29], Vygotsky's Zone of Proximal Development (ZPD) [30], Blair and Razza's neuropsychological evidence [18], and Van der Kolk's trauma neuroscience [11].
3. A Standardized Cognitive Content Layer delivering identical, curriculum-aligned cognitive activities to all children who pass the psycho-attentional gate. Content standardization at the delivery level, combined with individualized gating at the readiness level, enables valid cross-child performance comparison while respecting individual psychological differences consistent with Bloom's conditional standardization principle [29] and directly serving the NIE Sri Lanka Grade 3-4 curriculum framework [31].
4.

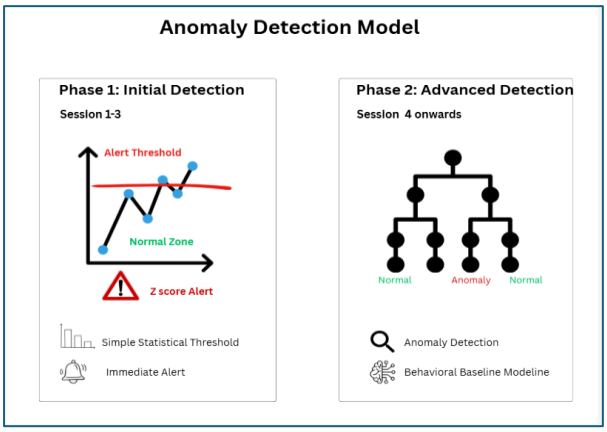


Figure 1: Anomaly Detection Model

A Machine Learning Anomaly Detection System (Figure 1) employing a Z-Score statistical method (Phase 1, operational from session three per child) progressing to an Isolation Forest model[32] (Phase 2, after sufficient session accumulation) that identifies children exhibiting persistent psycho-attentional distress across multiple sessions specifically three or more consecutive flagged sessions, or three of five recent sessions and generates plain-language, actionable welfare alerts for administrators via the synchronized cloud dashboard.

5. An Offline-First IoT Deployment Architecture in which the web application, local SQLite database, ML inference, and gating logic operate entirely on the Smart-Camp Box without internet dependency. The dedicated WiFi card creates a dual-mode access point (AP+STA), enabling simultaneous student device connections and internet detection. A Sync Model automatically

uploads session data, child profiles, and anomaly flags to cloud storage upon WiFi availability, enabling admin dashboard access, cloud-based model improvement, and updated model redeployment to the device.

1.1.6 SDG and Research Cluster Alignment

PAGES contributes to five of the United Nations' Sustainable Development Goals. The good health and well-being goal of SDG3 is addressed through the psychological stabilization function. The psycho-attentional monitoring and anomaly detection system provides continuous individual mental health surveillance for vulnerable children, generating welfare alerts that support Target 3.4 (mental health and well-being promotion). The quality education goal of SDG4 is addressed through the gate learning engine. Directly supports Target 4.1 (quality primary education completion) and 4.a (inclusive, safe learning environments). PAGES ensures Grade 3-4 children in disaster camps receive education only when neurologically ready, maximizing quality over quantity of instruction. The goal of SDG10, which addresses reduced inequality, is addressed through Offline-first IoT deployment, making psychologically aware educational technology accessible in under-resourced, disaster-affected communities without internet dependency, directly reducing educational equity gaps, Target 10.2. The peace, justice, and strong institutions' goal of SDG16 is Peace, Justice, and Strong Institutions. Supports development of resilient educational institutions functioning during crises-Target 16.6 and provides a child welfare monitoring infrastructure for displaced communities-Target 16.7.

PAGES finds alignment with the research cluster of the Centre of Excellence for AI (CEAI) in the following domains: The primary alignment is with Machine Learning (ML), which is achieved through the implementation of a two-phased anomaly detection mechanism, specific to every child, which uses Z-score statistical thresholding from session three onwards, followed by a scikit-learn library-based Isolation Forest unsupervised learning model from session eight onwards. The mechanism is used for ten behavioral features from every session, as there are no labeled ground truth values available. The secondary alignment is with Data Science (DS), which is achieved through the implementation of a multi-source behavioral signal fusion mechanism that normalizes, weights, and aggregates the metrics of the gamified tasks as well as the passive interaction features into a composite psycho-attentional score, which is updated through a rolling exponentially weighted baseline for every child over the sessions. The tertiary alignment is with Decision Support Systems, as the system can alert the welfare system through the alerts generated by the ML mechanism, which are used to notify the administrators for the allocation of volunteers for the various camps, as well as with Knowledge Management, as the behavioral profiles for every child are continuously updated and persisted across the edge and cloud layers, thereby reflecting the knowledge retention for every child, which is specific to the humanitarian IoT domain.

1.2 Literature Review

1.2.1 Trauma Neuroscience and the Neurobiological Basis of Learning Disruption

1.2.1.1 Van der Kolk - Trauma's Impact on the Learning Brain

The work by Van der Kolk, "The Body Keeps the Score" (2014) [11], constitutes the most comprehensively evidenced account of the neurobiological impact on children's cognitive function currently available in the literature. By integrating the results of decades-long research into the clinical and neuroimaging consequences of trauma, Van der Kolk shows that the sustained hyperactivation of the amygdala during post-traumatic stress results in the suppression of prefrontal cortex activity. This is the primary neurobiological rationale for the PAGES gating model. This is because, before any cognitive content is provided, the PAGES model must first ensure the psycho-attentional stability of the individual. This is because not only is the presentation of any academic content to the traumatized

individual irrelevant, but it can also re-traumatize the individual by associating the educational environment with the neurologically compromised state of failure. The primary limitation of Van der Kolk's work, as it pertains to the PAGES model, is the fact that the research is based almost exclusively on high-income populations and does not directly address the potential of technology-based behavioral signals to detect attenuated variants of these stress states. This is the purpose of the evaluation component of the current work.

1.2.1.2 Blair and Razza -Readiness as Biological Prerequisite

Blair and Razza's landmark study on effortful control and executive function in young children [18] established that psychological self-regulation functions as a prerequisite biological condition for learning, not merely a facilitating factor. Children who lack emotional and attentional self-regulation demonstrate significantly lower emergent literacy and numeracy skills, independent of curriculum quality or instruction time. This directly justifies the gating architecture of PAGES: confirming psychological and attentional readiness before cognitive engagement is not a pedagogical preference but a neurological necessity for Grades 3-4 children. While some affective computing systems attempt to detect stress or disengagement, PAGES is novel in explicitly operationalizing this principle as a computational content gate.

1.2.2 Educational Psychology Frameworks Underpinning the Gating Architecture

1.2.2.1 Maslow's Hierarchy of Needs



Figure 2: Maslow's Hierarchy of Needs

Maslow's motivational hierarchy [33] (Figure 2) provides the macro-level theoretical justification for PAGES's gating model. The hierarchy's central thesis is that higher-order cognitive and developmental needs cannot be adequately addressed until lower-order safety and belonging needs are met. PAGES operationalizes Maslow's principle of sequential prioritization of needs: a child does not receive cognitive content until behavioral signal data confirms that their psycho-attentional state has achieved a threshold level of stability. Academic content is delivered only when behavioral signals confirm sufficient psycho-attentional stability. The Cyclone Ditwah field documentation supports this framing, with UN SitRep data confirming that Maslow-informed sequential response protocols, physiological needs addressed first, psychological safety second, and academic education third, were identified as most effective across comparable disaster contexts, including Cyclone Idai, Mozambique, 2019 [34]. PAGES is, to the best of the authors' knowledge, a proposed computational implementation of Maslow's hierarchy as a dynamic content-access gate in an IoT-deployed educational technology system.

1.2.2.2 Vygotsky's Zone of Proximal Development

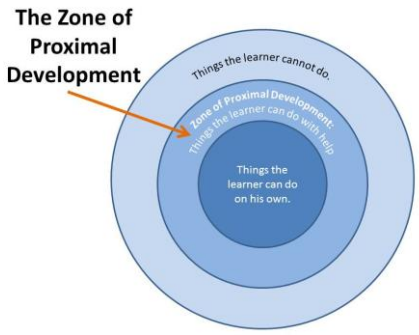


Figure 3: Vygotsky's Zone of Proximal Development

Vygotsky's Zone of Proximal Development [30] (Figure 3) defines the optimal learning zone as the space between independent performance capacity and the upper boundary of guided performance capacity. In the PAGES context, the ZPD is not used to calibrate content difficulty; it is standardized, not adapted, but rather to define the precondition for ZPD-aligned learning to be possible at all. A child in acute psychological distress is, neurologically, outside any productive ZPD: trauma-suppressed prefrontal function means that even appropriately levelled content cannot be processed. PAGES's gate operationalizes the ZPD's implicit precondition: a child must be in a neurologically viable state before the concept of 'just-right challenge' has any meaning. The Perplexity research synthesis (2026) [35] confirms that ZPD-aligned offline tools could extend kit-based learning in Sri Lankan camps. PAGES provides the psycho-attentional prerequisite infrastructure that makes such ZPD alignment educationally productive.

1.2.2.3 Bloom's Mastery Learning and Conditional Standardization

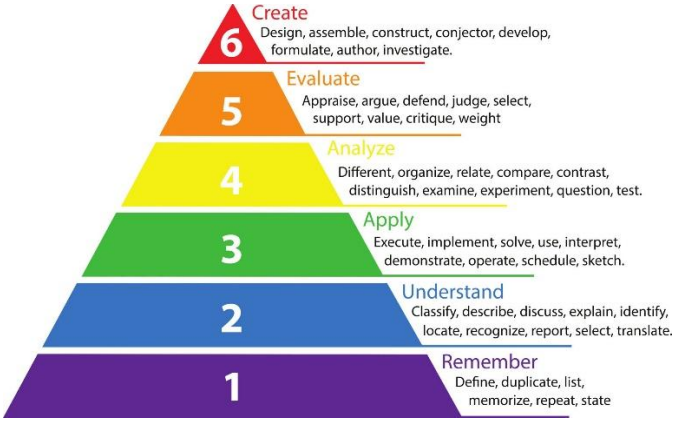


Figure 4: Bloom's Taxonomy of Learning Domain Levels

Bloom's Mastery Learning Model (1968) [29] (Figure 4) demonstrated that when students are brought to a sufficient readiness level before instruction, standardized content produces strong and equitable learning outcomes across diverse learners. This provides the theoretical basis for PAGES's standardized cognitive content layer: content uniformity is not a simplification, but a methodologically deliberate choice supported by Bloom's evidence, enabling meaningful cross-child performance comparison. The distinction between PAGES and adaptive learning systems is sharpened by Bloom's model: PAGES adapts the gate (individualized readiness assessment) rather than the content (standardized for all who pass). This is a theoretically coherent and evidence-grounded architectural choice, not a constraint.

1.2.3 Behavioral Assessment Methods for Children

1.2.3.1 Conners' Continuous Performance Test and Go/No-Go Paradigms

Conners' Continuous Performance Test (CPT, 1994) [27] is a widely used computerized measure of sustained attention in children. It was established that attention and psychological arousal in children can be reliably measured through behavioral response patterns, reaction time, error rate, and consistency without self-report. The CPT has been validated for children as young as six years old. Cragg and Nation (2008) [26] specifically studied Go/No-Go response inhibition tasks with children aged 6-11, confirming that response accuracy and reaction time in simple tap/don't-tap games are valid, age-appropriate measures of attentional control and psychological arousal in this population. The PAGES simple tasks are a child-friendly touchscreen implementation of this paradigm, yielding the same four measurement dimensions (accuracy, false responses, reaction time, and reaction time variance) in a format appropriate for Grade 3 children with no self-report requirement. Sergeant et al. (2000) [36] further confirmed that reaction time variability is the single most sensitive indicator of attentional state in children, a finding directly incorporated in the PAGES composite scoring algorithm.

1.2.3.2 D'Mello and Graesser - Multi-Signal Behavioral Fusion

D'Mello and Graesser (2012) [37] demonstrated that combining multiple behavioral signals produces significantly more accurate learner state detection than any single signal in isolation. Their multi-signal fusion approach to interacting data provides the methodological precedent for the PAGES three-source fusion model (50:30:20 weighting of gamified task, passive signals, and historical baseline). The 50% weight assigned to the gamified task reflects its status as the freshest, most direct behavioral measurement from the current session; the 30% weight for passive signals reflects continuous in-session behavioral monitoring; and the 20% historical baseline weight provides contextual stability against environmental noise, a critical consideration in the chaotic conditions of a disaster relief camp.

1.2.4 The Gating Model vs. Adaptive Learning

1.2.4.1 Corbett and Anderson - Knowledge Tracing and Its Limitations

Corbett and Anderson's (1994) Knowledge Tracing model [21], the foundational architecture for adaptive tutoring systems, adjusts content difficulty based on observable performance data, including correct and incorrect responses. Modern adaptive learning systems, including Deep Knowledge Tracing (Piech et al., 2015) and commercial platforms including DreamBox and Knewton, operate within this performance-based adaptation paradigm. The critical limitation of this paradigm in the post-disaster context is its universal assumption that the learner is cognitively ready to engage and that performance variability reflects differences in domain knowledge rather than psychological or attentional state. In disaster-affected Grade 3-4 children, this assumption is specifically and systematically violated: poor performance is as likely to reflect trauma-induced cognitive suppression as genuine academic knowledge gaps. PAGES addresses this by introducing a pre-learning state assessment and gating layer that performance-based adaptive systems largely omit, not by adapting content difficulty, but by confirming readiness before any content is delivered.

1.2.5 Offline IoT-Based Educational Technology

1.2.5.1 Kolibri and Raspberry Pi Deployment - Precedent and Limitations

Kolibri, documented by the Organisation for Economic Co-operation and Development (OECD) (2021) [24] for deployment in Jordan refugee camps using Raspberry Pi servers, provides a directly relevant existing system to PAGES's deployment architecture. Kolibri's widespread use in over 200 countries demonstrates the feasibility of the Raspberry Pi offline server model for humanitarian

contexts. However, the OECD evaluation explicitly noted that Kolibri's absence of adaptive personalization limited its effectiveness for heterogeneous camp populations. PAGES extends significantly beyond Kolibri by adding a psycho-attentional assessment layer, a gating model, anomaly detection for welfare monitoring, and automatic cloud synchronization, transforming a static offline content library model into a psychologically aware, welfare-monitoring educational system.

1.2.6 Machine Learning Anomaly Detection for Child Welfare Monitoring

1.2.6.1 Isolation Forest Unsupervised Per-Child Anomaly Detection

Liu et al.'s (2008) Isolation Forest algorithm [36] is the proposed Phase 2 anomaly detection model for PAGES, as shown in Figure 1. Isolation Forest operates unsupervised, requiring no pre-labeled 'abnormal' examples, making it ideal for the PAGES context where labeled instances of child psychological distress patterns are unavailable. It isolates anomalies through random feature partitioning: anomalies, being statistically rare and behaviorally different from the child's personal baseline, require fewer partitions to isolate than normal observations. Chandola et al.'s (2009) comprehensive survey [38] confirms that Isolation Forest-type approaches are most appropriate when individual behavioral baselines vary significantly across subjects and labeled anomalies are scarce, exactly the PAGES deployment condition. The two-phase architecture Z-Score statistical detection from session three onwards (Phase 1) transitioning to Isolation Forest after sufficient data accumulation (Phase 2) mirrors common practice in patient healthcare monitoring, where simple statistical thresholds provide immediate alerting while more sophisticated models are trained on accumulating patient data [38]. This phased approach ensures PAGES is functional from the first sessions while progressively improving accuracy over time.

1.2.7 Post-Disaster Education: Systematic Review Evidence

Crompton et al.'s (2021) systematic review [21], examining 60 studies across 48 countries between 2010 and 2020, represents one of the most comprehensive systematic reviews to date on educational technology in crisis contexts. The review found that while technology-assisted education in emergencies produced generally positive engagement outcomes, adaptive platforms were significantly underrepresented in the evidence base, and internet-based tools systematically excluded the most vulnerable learner populations in low-connectivity settings. The review highlights the lack of offline-capable adaptive tools as a priority research gap, a gap that PAGES's gating model and offline-first IoT architecture directly addresses. The review's finding that 'personalized digital learning produced engagement gains even in low-resource settings' is consistent with the PAGES approach, while the caveat that tools requiring internet connectivity 'systematically excluded the most vulnerable' directly motivates the offline-first architectural commitment.

1.3 Research Gap

1.3.1 Identification of the Research Gap

The literature review section identifies a research gap that is both specific and significant.

1.3.1.1 No Pre-Learning Psycho-Attentional Gating in Any Existing Educational Technology

All existing educational technology systems, from Knowledge Tracing [21] to commercial-oriented adaptive systems, share the fundamental assumption that the student is ready to learn and that the adaptation of the content is a reaction to the student's performance indicators after the learning. Currently, no educational technology system assesses, before the start of the learning process, whether the child is psychologically and attentively ready to learn any content at all. As shown by the neuroscientific evidence presented in Sections 1.1.1 and 2.1 above, the assumption is explicitly false

for disaster-affected children between the ages of 7 and 9, where psychological distress physiologically inhibits the cognitive capacity for learning [11][3]. An educational content delivery system for a psychologically distressed child is not only useless; it is also a form of pedagogical harm because it requires the cognitive performance of a child who is neurologically incapable of delivering it. This paper presents the first documented proposal for the operationalization of the readiness gates before the learning process for a deployable educational technology system for young disaster-affected children.

1.3.1.2 No Behaviorally Indirect Psycho-Attentional Assessment for Non-Self-Reporting Children in Educational Technology

Existing educational technology systems for learner assessment mainly focus on facial expression-based assessment using webcams, physiological sensors, and direct self-report questionnaires [37], which are impossible for Grade 3-4 children in a disaster camp environment. Using webcams and physiological sensors requires costly hardware that cannot be utilized with the Smart-Camp Box. Facial expression-based assessment using images and videos of children poses critical issues regarding child data privacy. Direct self-reporting is not suitable for children below the age of 9, who lack introspection to accurately report their own states of mind [28]. The PAGES method, using a gamified Go/No-Go behavioral task [26], passive interaction signals [27], and a pre-computed rolling historical baseline, represents a methodologically unique and hardware-minimal solution for psycho-attentional state assessment in young non-self-reporting children. This unique combination of assessment sources for this specific learner population is not found in any other published system.

1.3.1.3 No Anomaly Detection for Child Welfare Monitoring in Educational IoT Systems

Existing educational IoT and EdTech systems invariably produce learner analytics focused only on academic performance metrics such as assessment scores, time spent on tasks, topic mastery, and learning curves. Currently, no such system uses anomaly detection techniques to detect children who display patterns of psychological distress across sessions and produce actionable welfare notifications for physical intervention. This is especially relevant in disaster scenarios. It is documented by Save the Children that tens of thousands of children in temporary shelters need mental health support [25], with a significant lack of qualified counsellors. The anomaly detection system developed for the PAGES project is especially relevant for such a disaster scenario, where the anomaly detection system enables non-specialist admin who do not have psychological training to receive notifications about children who display patterns of psychological distress based on the data collected from the children. This human-in-the-loop welfare monitoring architecture has no precedent in existing educational technology literature.

1.3.1.4 No Offline-First IoT System Combining Psycho-Attentional Gating and Welfare Monitoring for Post-Disaster Deployment

Although the technical feasibility of offline content delivery for the Raspberry Pi-based system was shown in the context of the Kolibri offline content delivery system [24], and despite the presence of IoT-based educational systems for well-resourced contexts, no system currently exists that integrates all of the following within a single, deployable component for use in post-disaster contexts: (a) an offline-based IoT system, (b) pre-learning assessment of psycho-attentional states through behaviorally indirect means, (c) control of cognitive content access based on learner readiness, (d) machine learning-based anomaly detection for welfare, and (e) automatic cloud synchronization for the persistence of data and model updates. This was explicitly identified as a research priority for the educational technology community, as noted by Crompton et al. (2021) [21]. In fact, the Rapid Evidence Review conducted by the EdTech Hub found that offline adaptive tools can help to mitigate infrastructure and

equity barriers for crisis contexts, though it found that the overall evidence for such tools was scarce. Accordingly, the PAGES system addresses the research priority.

1.3.2 Why Current Approaches Are Insufficient

1.3.2.1 Technological Insufficiency

The educational platforms that are currently available are designed to work with consistent connectivity, hardware availability, and educated instructors. They are not designed to run on a portable IoT device with low connectivity, operated by a admin with no technical expertise, and accessed by children on shared devices in an overcrowded disaster camp. The offline-focused, IoT-based deployment model of PAGES fills the technological gap.

1.3.2.2 Architectural Insufficiency

Existing adaptive systems change the content of instruction based on a child's performance. PAGES, however, poses a fundamental, architecturally prior question: is instruction appropriate at a given time for a particular child? This is a paradigm shift, not a variation of existing adaptive learning systems, and is supported by a body of research in developmental psychology and trauma neuroscience, which existing systems have neither attempted nor validated. The lack of a gating layer is not merely a problem to be solved but a fundamental architectural choice based on the belief that a learner is always ready to learn. PAGES systematically challenge and computationally answer this assumption.

1.3.2.3 Humanitarian Insufficiency

Existing systems provide learner analytics to trained educators in a predictable institutional environment. In contrast, PAGES provides welfare alerts to admins in chaotic disaster environments, where a child's psychological well-being may have greater immediate implications for their welfare than their academic achievement. There is no existing educational technology system with this humanitarian priority structure, in which monitoring child welfare is an equal to or greater priority than providing educational content.

1.3.3 Justification of Novelty and Research Contribution

The application of pre-learning psycho-attentional gating based on Piaget's stages of cognitive development [5], Maslow's hierarchy of needs [33], Bloom's mastery learning theory [29], Vygotsky's Zone of Proximal Development [30], and the neuroscientific studies by Van der Kolk, Blair, and Razza [11][3], and packaged into a deployable form of educational technology, is a theoretically novel contribution to the fields of Information Technology Services (ITS) and affective computing.

The novel approach to behavioral fusion from three sources (gamified Go/No-Go task, passive interaction signals, and rolling historical baseline) for psycho-attentional measurement is a methodologically novel approach for the measurement and assessment of psycho-attentional states without the need for direct self-reporting from children, with no direct precedent in published research on educational technology.

The offline Internet of Things (IoT) architecture combining machine learning scoring, behavioral measurement, Isolation Forest anomaly detection, and automatic cloud synchronization into a single portable device is a technically novel approach to the deployment of Internet of Things (IoT) architecture for the deployment of AI in education.

The post-disaster primary education context in Sri Lanka, supported by empirical data from Cyclone Ditwah referenced by [1][2], is a contextually novel and underserved context for the deployment of AI in education, which existing published research on educational technology has not satisfactorily addressed. This is a research gap identified by Crompton et al. [21].

1.3.4 Evidence-Based Argument for Research Necessity

Empirical Scale: The Cyclone Ditwah (2025) disaster affected 890,000 children, destroyed 1,682 schools, resulted in a teacher-to-student ratio of 1:77, and resulted in tens of thousands of children in relief camps who require both educational support and mental health care, with zero AI-driven educational systems in place to support the response efforts [1][2].

Neuropsychological: The psychological distress of children in the 7-9 year age group neurologically inhibits the prefrontal cognitive functions necessary for learning, making it both ineffective and counterproductive to educate children in a distressed psychological state [3][18].

Methodological: The behavioral indirect methods of Go/No-Go Gamified Tasks and Passive Interaction Analysis offer a reliable and valid method of measurement of the psychological and attentional state of children in the 6 to 11 year age group, without the requirement of any specialized hardware or self-report, and have been supported through several peer-reviewed studies [26][27][37].

Technical: The technical viability of an offline-first IoT-based educational system using Raspberry Pi hardware in a post-disaster setting, while educationally effective in settings such as the Jordanian Refugee Camps with Kolibri, does not exist when incorporating a psycho-attentional gating mechanism or welfare anomaly detection in a low-connectivity network.

Literature Gap: No current educational technology exists that incorporates a psycho-attentional pre-learning gating mechanism, behavioral state measurement, machine learning-based anomaly detection for welfare monitoring, and IoT deployment in a single component for post-disaster primary education support, low connectivity, as evidenced in the literature [21][38].

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate PAGES as an offline-first, IoT-deployed web application component of the Smart-Camp Box framework function in low network connectivity and that assesses the combined psycho-attentional readiness state of Grade 3 and Grade 4 children (ages 7-9) in post-disaster Sri Lankan relief camps through indirect behavioral methods, gates access to standardized cognitive educational content based on quantified readiness thresholds, and employs machine learning anomaly detection to identify children exhibiting persistent psychological distress and generate welfare alerts for admin and synchronizing session data to cloud infrastructure upon connectivity availability.

2.2 Specific Objectives

The five specific objectives that follow are each measurably specific, sufficient in total to achieve the general objective, and consistent with the research gap identified in this thesis, the PAGES system architecture, and the Smart-Camp Box model in general.

2.2.1 Design and Validate Psycho-Attentional Assessment Layer

The objective here is to develop and validate an integrated psycho-attentional assessment layer that measures the psychological and attentional readiness state of Grade 3-4 children through three types of indirect behavioral data sources, without directly asking the child any questions. The proposed assessment is as follows: (i) a gamified Go/No-Go Paradigm Task, which measures accuracy rates, false tap rates, mean reaction time, and reaction time variance (50% weight); (ii) passive behavioral interaction signals, which measure idle time ratios, rapid click counts, time to first interaction, and completion ratios (30% weight); and (iii) a pre-computed rolling baseline score, which is computed

using a weighted rolling average score for each session, defined as New Baseline = Old Baseline x 0.8 + Current Session Score x 0.2 (20% weight). The final score, computed by combining all three data sources, ranges from 0 to 100 and is to be validated.

2.2.2 Implement and Evaluate the Psycho-Attentional Gating Model

To implement and evaluate a psycho-attentional gating model that applies the composite score from 2.2.1 to determine, per session and per child, whether access to the cognitive content layer is granted or withheld delivering a brief calming or re-engagement support activity to children below the readiness threshold, re-assessing their state, and granting access upon confirmation of sufficient readiness, thereby ensuring cognitive content is never delivered to a child in a neurologically non-viable learning state.

2.2.3 Develop a Standardized Cognitive Content Layer for Grade 3-4 Children

To develop a standardized cognitive content layer delivering uniform, curriculum-aligned learning activities to all children who pass the psycho-attentional gate, aligned with the National Institute of Education, Sri Lanka Grade 3-4 curriculum framework [31], and to evaluate the layer's educational efficacy by measuring cognitive performance scores and session progression rates across the study population.

2.2.4 Develop and Evaluate a Machine Learning Anomaly Detection System for Child Welfare Monitoring

The study aims to develop and evaluate a Machine Learning-based anomaly detection system that detects children's psycho-attentional distress. This is a two-phase Machine Learning-based anomaly detection system. In Phase 1, a Z-Score-based statistical thresholding technique is applied starting from the third session. Once a substantial amount of session data is collected, Phase 2 of the system is activated, using the Isolation Forest Machine Learning technique. Children are identified as distressed if they have a series of sessions that are below the threshold, which is three or more consecutively, or if they have had three of the five most recent sessions that are below the threshold. Welfare alerts are then generated and sent to online administrators through a dashboard.

2.2.5 Deploy and Validate the Offline-First IoT Architecture with Synchronization Model

To deploy and validate the complete PAGES system on Smart-Camp Box hardware a Raspberry Pi 5-based edge computing device demonstrating full functional operation of the web application, local SQLite storage, psycho-attentional assessment, gating model, cognitive content delivery, and ML anomaly detection without any internet connectivity, and to validate the WiFi card dual-mode (AP+STA) architecture for simultaneous child device connections and internet detection, and the Sync Model for automatic cloud data synchronization upon connectivity availability.

3. METHODOLOGY

3.1 Research Approach

3.1.1 Research Paradigm

This research adopts a Design Science Research (DSR) paradigm as its primary methodological framework. DSR is appropriate when the goal is to design, build, and evaluate an artefact that solves a real-world problem which precisely describes the PAGES component. The artefact in this study is the PAGES system itself: a working software system that integrates psycho-attentional assessment, gating logic, standardized cognitive content, machine learning anomaly detection, and offline IoT deployment.

Following Hevner et al. (2004), the research proceeds through three design science cycles: the relevance cycle (grounding the design in the real-world problem of post-disaster primary education), the design cycle (iterative design, build, and evaluation of PAGES), and the rigour cycle (grounding design decisions in existing knowledge from trauma neuroscience, behavioral assessment, and anomaly detection literature).

Within the DSR framework, the study incorporates an experimental evaluation component for pilot validation. A within-subject quasi-experimental design is used during the pilot phase, in which each child serves as their own control across session conditions, enabling valid comparison of system behavior under varying attentional states without requiring a separate control group.

3.1.2 Research Design Overview

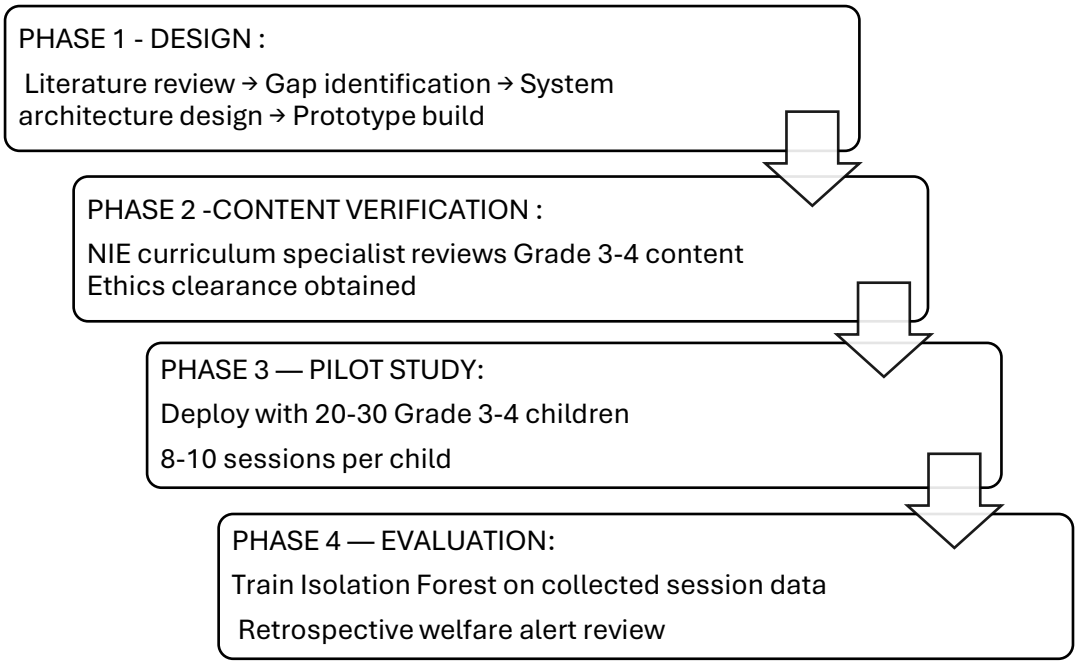


Figure 5: Research Design Overview

3.1.3 Rationale for Participant Population and Context

Due to the temporal distance from the acute phase of Cyclone Ditwah (approximately four months at the time of the pilot study), acutely displaced children in active relief camps were not available to participate. To generate the attentional variation necessary for meaningful evaluation of the psycho-attentional assessment layer and the anomaly detection system, a within-subjects experimental design was employed. Sessions were conducted under three conditions: a baseline condition in which children attended normally, a cognitive load condition in which children completed a brief working memory task immediately before the PAGES session, and a fatigue condition in which sessions were scheduled at the end of the school day. This design is grounded in established cognitive load theory (Baddeley 1992) and validated Go/No-Go attention research (Cragg and Nation 2008), both of which document attentional performance degradation under equivalent conditions in children of this age. It is acknowledged that cognitively induced attentional reduction is a proxy for the trauma-induced neurological suppression documented in post-disaster contexts by Van der Kolk (2014) and Blair and Razza (2007). Evaluation of PAGES under conditions of genuine acute psychological distress is identified as a priority for future research.

3.2 Technical Methods and Algorithms

3.2.1 Combined Psycho-Attentional Assessment

The psycho-attentional assessment layer fuses three behavioral data sources into a single composite score (0-100) using a weighted fusion formula:

Composite Score = (Gamified Task Score × 0.50) + (Passive Signal Score × 0.30) + (Rolling Baseline × 0.20)

Source 1 - Gamified Go/No-Go Task (50% weight)

A 90-second touchscreen task based on the Go/No-Go inhibitory control paradigm validated by Cragg and Nation (2008) [26] for children aged 6-11. The child taps target animals (Go stimuli) and suppresses responses to distractor animals (No-Go stimuli). Four metrics extracted are in Table 3.

Table 3: Gamified metrics

Metric	Formula	What It Measures
Accuracy	RateCorrect taps ÷ Total target appearances	Sustained attention and task engagement
False Tap Rate	Distractor taps ÷ Total distractors	Impulse control is elevated under distress
Average Reaction Time (ms)	Mean of valid tap response times	Cognitive processing speed
Reaction Time Variance	Standard deviation of reaction times	Attentional consistency most sensitive indicator (Sergeant et al. 2000)

Source 2 - Passive Behavioral Interaction Signals (30% weight)

Four signals (Table 4) collected continuously from the child's browser session without any additional input:

Table 4: Passive behavioral interaction signals

Signal	Definition	Distress Indicator
Idle Time Ratio	Seconds with no interaction ÷ Session duration	High idle = disengagement
Rapid Click Count	Taps above the 3 per second threshold	Elevated = anxiety or frustration
Time to First Interaction (ms)	Duration from page load to first tap	Long delay = hesitancy or withdrawal
Session Completion Ratio	Screens completed ÷ Total screens	Low = inability to persist

Source 3 - Rolling Historical Baseline (20% weight)

A pre-computed baseline score for each child, updated after every session:

New Baseline = (Previous Baseline × 0.8) + (Current Score × 0.2)

0.8/0.2 weighting applies a smoothing decay that gives more weight to the child's established history than to any single session, preventing one anomalous session from distorting the baseline.

3.2.2 Psycho-Attentional Gating Algorithm

The gating model applies a binary threshold decision per session per child:

- If Composite Score ≥ Threshold:

  GATE = PASS

  - deliver cognitive content

- If Composite Score < Threshold:

  GATE = FAIL

  - delivering calm support activity
  - reassess after activity
  - second gate decision
  - if still FAIL: mark session,
  - increment consecutive flag counter

The readiness threshold is initialised at 55/100 based on the midpoint of the moderate readiness range and is subject to calibration during pilot evaluation. The threshold is applied uniformly; it does not adjust per child. What adjusts per child is the rolling baseline that forms 20% of the composite score, providing individualisation at the input layer rather than the decision layer.

3.2.3 Machine Learning Anomaly Detection - Two-Phase Architecture

Phase 1 - Z-Score Statistical Method (Session 3 onwards)

The Z-Score method requires no training data and activates as soon as three sessions exist for a child:

Z = (Current Composite Score − Child's Mean Score) ÷ Child's Standard Deviation

Threshold: Z < -2.0 → session flagged

Interpretation:

Z = -1.0 → slightly below average (no flag)

Z = -2.0 → bottom 2.3% of normal distribution (flagged)

Z = -3.0 → extreme deviation (flagged, high concern)

The child's mean and standard deviation are recalculated after each session using the entire session history, making the method increasingly accurate as sessions accumulate.

Phase 2 - Isolation Forest (Session 8 onwards)

Isolation Forest (Liu et al. 2008) [32] is an unsupervised anomaly detection algorithm that identifies anomalies by measuring how easily a data point can be isolated from the rest using random decision

trees. Anomalous sessions, being rare and behaviorally distinct from the child's normal pattern, require fewer splits to isolate and therefore receive lower anomaly scores.

3.2.4 Rolling Baseline Update Algorithm

def update_baseline (old_baseline, current_score, alpha=0.8):

Exponentially weighted moving average.

alpha = weight given to historical baseline.

(1 - alpha) = weight given to current session.

new_baseline = (old_baseline * alpha) + \ (current_score * (1 - alpha))

return round(new_baseline, 2)

The alpha value of 0.8 was selected to provide a stable baseline that responds gradually to genuine improvement while remaining resistant to single-session noise, consistent with the exponential smoothing literature for short time-series data.

3.3 Tools and Platforms with Justification

3.3.1 Technology Stack Overview

- Frontend: Next.js
- Backend: Node.js + Python Flask (ML inference API)
- Edge Database: SQLite (offline edge device)
- Cloud Database: AWS DynamoDB
- ML engine: Python + scikit-learn
- Hardware: Raspberry Pi 5
- Power System - Solar Panel with Lithium Battery Pack
- Hosting: Vercel (admin dashboard) + Supabase (cloud sync storage)
- Version control: Git + GitHub IDE Visual Studio Code
- UI design: Figma

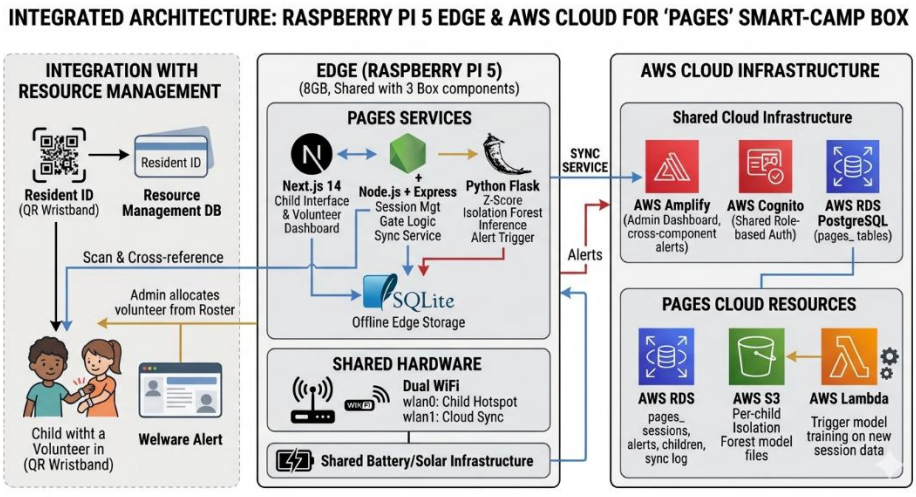


Figure 6: Integrated Architecture

3.3.2 Justification for Each Tool

Next.js- Child-Facing and Admin Interfaces

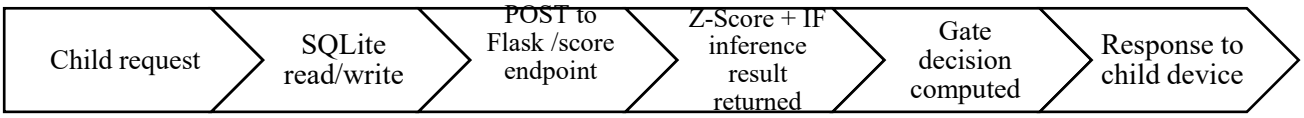
Next.js enhances the features of the React library and provides server-side rendering and static site generation features that are directly advantageous within the PAGES deployment context. Server-side rendering allows the HTML of the pages to be rendered on the Raspberry Pi server before it is sent to client devices. This is advantageous because it reduces the workload on the client devices, which might be of low specification and might not be able to render the pages quickly. This advantage is significant because the Raspberry Pi server will be handling the workload of the client devices. Since the Raspberry Pi server will be handling the workload of the client devices, it will be advantageous if fifteen or more child devices are connected simultaneously to the Smart-Camp Box access point. The file-based routing of the Next.js framework will be advantageous because it will allow the clear separation of routes for the child devices, the routes for the volunteer dashboard, and the API routes within the same application.

Node.js - Primary Backend API

The combination of Node.js serves as the primary REST API for managing child sessions, retrieving child profiles, routing gate decisions, and content delivery. The non-blocking I/O mechanism provided by Node.js is well-suited to managing multiple simultaneous connections from child devices on the Raspberry Pi 5 without blocking, thereby meeting the requirement of 15 or more concurrent users with a response time of less than 3 seconds.

Python + Flask - Machine Learning Inference API

A separate Python Flask microservice runs alongside the Node.js backend to support ML operations, including Z-score calculation, Isolation Forest inference, rolling baseline updates, and alert-trigger evaluation. Python is the only practical choice for this layer given scikit-learn's Isolation Forest implementation, NumPy's statistical functions, and the absence of equivalent production-grade ML libraries in the Node.js ecosystem. Flask is selected over Django for its minimal footprint on Raspberry Pi 5 hardware; avoiding unnecessary framework overhead is operationally important. The two backends communicate internally:



SQLite - Offline Edge Database

SQLite was chosen for the Raspberry Pi installation based on four context-specific criteria. SQLite eliminates the need for a separate server process and therefore requires no additional RAM on the Raspberry Pi. SQLite stores the entire database in a single file on the microSD card, making backups and recovery easy in the event of a power failure. SQLite is fully ACID-compliant, ensuring data integrity even in the event of a power failure. SQLite can also handle the projected data volume (over 1,000 session records) with millisecond read/write latency on Raspberry Pi 5 hardware. No other embedded database system offers maturity, toolset support, and zero-configuration requirements.

AWS DynamoDB - Cloud Layer Database

DynamoDB is a fully managed serverless NoSQL document database that eliminates database management. The document-oriented nature of DynamoDB is also noteworthy; it can handle the dynamic nature of session records stored in PAGES. The session record has a constant set of basic information along with a variable-length sub-document that represents behavioral signals. The sub-document has a dynamic structure that could change as the system is improved upon. Such a dynamic structure is difficult to achieve in a relational database without a high cost. Moreover, DynamoDB is native to the AWS ecosystem that is already being utilized at various levels within the PAGES cloud layer; thus, authentication is not a problem.

Raspberry Pi 5- Edge Computing Hardware

Raspberry Pi 5 is selected as the Smart-Camp Box hardware platform based on three validated precedents and four technical requirements. Validated precedents: Kolibri offline educational deployment in Jordan refugee camps (OECD 2021) [24], general IoT educational deployments documented in Crompton et al. (2021) [21], and the broader Smart-Camp Box framework selection. Technical requirements: sufficient RAM to run Node.js, Flask, and SQLite simultaneously; dual-band WiFi for AP+STA mode operation; USB 3.0 for reliable WiFi adapter connection; and 5V power input compatible with LiFePO4 battery output.

scikit-learn - Isolation Forest Implementation

Scikit-learn provides the most mature, extensively validated, and computationally efficient Python implementation of Isolation Forest, with direct support for the contamination parameter, score samples inference, and model serialization via joblib. The library's Raspberry Pi ARM architecture compatibility is confirmed through community deployment records.

Figma - UI Prototyping

Figma is used for child-facing interface mockups and admin dashboard wireframes during the design phase. The free tier is sufficient for the research prototype scope. Figma's collaborative features allow supervisor review of UI designs without software installation.

3.4 Data Collection and Ethical Considerations

3.4.1 Participant Profile

Table 5: Participant Profile

Criterion	Specification
Age	7-9 years (Grade 3 and Grade 4)
Location	Mullegama Maha Vidyalaya and Jalthara Maha Vidyalaya
Background	Children affected by or recovering from Cyclone Ditwah (November 2025) |
Number	30 children who are affected by Ditwah
Consent	Written parental/guardian consent is required before participation
Voluntariness	Participation entirely voluntary; withdrawal at any time without consequence

3.4.2 ZEO Permission and Content Verification Protocol

The Zonal Education Division, Homagama, granted permission for the pilot study. The ZEO has set the following condition before children interact with the system: The cognitive content of PAGES must be reviewed and verified as appropriate for Grade 3 and Grade 4 students. Verification process:

- Step 1: PAGES cognitive content layer built (literacy, numeracy, comprehension activities)
- Step 2: Content submitted to ZEO-designated reviewer for alignment review against NIE Sri Lanka Grade 3-4 curriculum framework
- Step 3: Reviewer confirms content is appropriate and provides written sign-off
- Step 4: ZEO grants final permission for child participant engagement
- Step 5: Parental consent forms distributed through the school
- Step 6: Pilot study begins

3.4.3 Data Collection Procedure

Each session is conducted in a supervised classroom setting. A facilitating teacher is present throughout each session block. Children access the PAGES application through a tablet browser connected to the Smart-Camp Box WiFi hotspot - no application installation is required. Sessions are conducted in groups of 5-8 children, each on an individual device.

- Session Structure (15–20 minutes):
    1. Teacher introduces the session (2 min)
    2. Child opens browser, enters child ID code (1 min)
    3. Gamified Go/No-Go task - Tap the Animal (90 sec)
    4. Gate decision displayed (automatic, < 5 sec)
    5. Session ends - data saved automatically
    6. Teacher completes brief session log (2 min)

3.4.4 Ethical Considerations

University Ethics Clearance

A full ethics application was submitted to the University Research Ethics Committee. The application will include a research summary, participant information, and what data will be gathered.

Child Data Protection

At no point is the actual child's name stored in the PAGES system. Identification is achieved through anonymous codes (for example, HMG-001), which are used at registration. The relationship between the actual identities and the anonymous codes is maintained in paper form only by the class teacher and not in any digital format. Session data is stored locally on the Raspberry Pi with password protection and is transmitted to cloud storage using HTTPS with TLS 1.2+ encryption.

Minimizing Psychological Risk

The gamified task has been designed to focus on the aspect of fun rather than difficulty, and the Go/No-Go paradigm is non-stressful for children between 6-11 years of age, according to Cragg & Nation (2008) [26]. The soothing activity provided upon the failure of the gate has been designed to be non-corrective and does not convey any negative message to the child. Children who are constantly failing the gate are not penalized but are provided with soothing activity and are still positively engaged by the system.

Right to Withdraw

Any child may stop at any time without any consequence. The class teacher is empowered to remove any child from a session at their discretion. Parents may withdraw their child from the study at any time and request deletion of their child's data.

3.5 Validation Strategy and Performance Metrics

3.5.1 Overview

Validation is structured around the five Specific Objectives defined in Section 2.2. Each objective has defined measurability targets that constitute pass/fail criteria for the evaluation.

3.5.2 Assessment Layer Validation

Validate: Does the composite psycho-attentional score reliably detect differences in attentional state for the same child?

Table 6: Assessment layer validation

Metric	Target	Method
Within-child score sensitivity	Mean stressor score significantly lower than baseline ($p < 0.05$)	Paired t-test
Gamified task consistency	Test-retest reliability $\alpha \geq 0.70$ across baseline sessions	Cronbach's alpha
Composite score range utilisation	Scores span at least 30 points across all sessions	Descriptive statistics

Justification for revised approach: The original target of $r \geq 0.65$ correlation with volunteer ratings was revised to a within-subject experimental comparison following recognition that per-session volunteer rating at scale is not practically feasible with the available facilitator capacity. The within-subject design provides equivalent statistical validity for detecting assessment sensitivity, as it controls all between-child confounds by using each child as their own control.

3.5.3 Gating Model Validation

Validate: Does the gate correctly distinguish between ready and less-ready states, and does the support activity help children recover?

Table 7: Gating model validation summary

Metric	Target	Method
Gate pass rate differential	Significantly higher pass rate in baseline vs stressor sessions ($p < 0.05$)	Chi-square test
False positive rate	$\leq$ 15% of stressor sessions incorrectly receive Pass	Condition-tagged session analysis
Session recovery rate	$\geq$ 60% of Fail-first sessions pass on the second attempt	Gate decision log analysis

3.5.4 Cognitive Content Layer Validation

Validate: Is the content curriculum aligned, and do children improve over sessions?

Table 8: Cognitive content layer validation summary

Metric	Target	Method
Curriculum alignment	≥ 90% of activities rated as aligned by the ZEO reviewer	Content review checklist
Performance progression	Statistically significant improvement across a minimum of 5 sessions ($p < 0.05$)	Linear regression on session scores per child
Session completion rate	≥ 80% of gate-passing children complete the full content session	Session completion ratio from SQLite

3.5.5 Anomaly Detection Validation

Validate: Does the system correctly identify children showing persistent attention distress and generate timely welfare alerts?

Table 9: Anomaly detection validation summary

Metric	Target	Method
Alert precision	≥ 0.75 (75% of alerts confirmed as warranted)	Retrospective teacher review
Alert recall	≥ 0.70 (70% of struggling children flagged)	Teacher-identified missed cases
Alert latency	≤ 3 sessions from pattern onset to alert generation	Session log timestamp analysis
Z-score vs. if agreement	≥ 80% agreement on flagged sessions where both phases are active	Cross-method flag comparison

3.5.6 IoT Architecture Validation

Validate**:** Does the complete system work offline under realistic deployment conditions?

Method: Technical system testing.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Diagram of Component

Figure 5 illustrates the high-level system architecture of the PAGES system, showing the three architectural layers, the five key components of the edge layer, the data flows between components, and the connectivity boundary between offline-first operation of the edge layer and cloud-based operations. The system has been designed with multiple layers such that all consumer-facing and safety-critical operations occur within the edge layer, even in the absence of connectivity, while the cloud-based operations provide enhancements and oversight when connectivity is available.

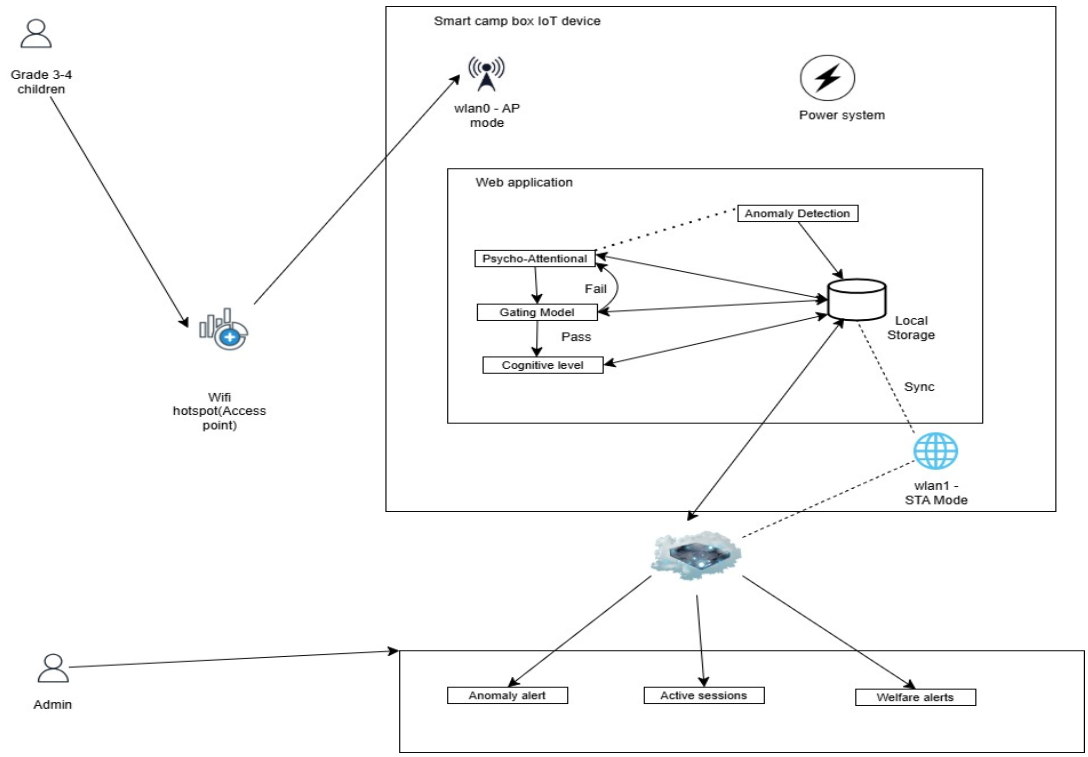


Figure 7: PAGES architectural diagram

4.2 Data Flow Explanation

Table 10: Data flow summary

Task	**Source**	**Destination**	**Data content**	**Condition**
Child interaction data	Child device (browser)	PAGES App (wlan0)	Tap events, timestamps, and accuracy signals from the gamified task	During every session, a continuous stream
Composite score	Assessment Layer (C1)	Gate Model (C2)	10-feature behavioral vector, composite score (0-100)	Immediately after the gamified task completes
Gate decision	Gate Model (C2)	Child device	Pass or fail decision; support activity content if fail	Within 5 seconds of composite score computation
Cognitive content	PAGES App	Child device	Curriculum-aligned learning activities, instructions, tasks	Only on the pass gate decision
Performance result	Child device	SQLite (sessions table)	Score, completion status, duration, performance record	After the cognitive session ends

Anomaly check input	SQLite (sessions table)	Anomaly Engine (C4)	Child session history, current 10-feature vector	After every session completes
Welfare alert	Anomaly Engine (C4)	Admin Dashboard	Child ID, alert type, plain-language message, suggested action	When 3 consecutive or 3/5 session flags are triggered
Baseline update	Session result	Child profile (SQLite)	New rolling baseline = (old × 0.8) + (current × 0.2)	After every session, automatic
Cloud sync-outbound	Smart-Camp Box (wlan1)	Cloud Database	Pending session records, anomaly flags, alerts, and child profiles	When wlan1 detects internet connectivity
Model update-inbound	Cloud ML System	Smart-Camp Box	Retrained Isolation Forest model files (.pkl) per child; content library updates	After cloud ML training completes on synced data

4.3 Scalability and Security Considerations

4.3.1 Security Considerations

PAGES operates in a context involving behaviorally sensitive data from vulnerable children in a humanitarian setting, requiring security measures appropriate to both the sensitivity of the data and the resource constraints of the deployment environment. Five security principles govern system design.

Child Data Anonymization and Two-Layer Identity Architecture

The child identity management in PAGES is designed to strictly separate child identity information from session information. This is achieved by a two-layer architecture, which interfaces with the Resource Management component of the Smart-Camp Box framework.

When a new resident is registered in the camp, the Resource Management component issues a physical wristband for each resident containing a unique QR code with a Resident ID, for example, RES-2847. The Resource Management database is the sole repository for each resident's identity information, including full name, age, grade, family linkage, camp zone, tent locations, guardian contact information, and medical information. This information is never duplicated in the PAGES database.

When a child is enrolled in the PAGES educational program, the child's wristband with a QR code containing the Resident ID is used through the PAGES registration interface. PAGES reads the Resident ID from the QR code and creates a new anonymized PAGES child ID, for example, HMG-012. This new anonymized child ID is stored in the PAGES database in a cross-reference field with the Resident ID, grade level, and the camp ID. All session information, composite scores, gate information, behavioral signals, flags, and welfare information are stored under the anonymized PAGES child ID in the local SQLite database on the Raspberry Pi device and in the shared AWS RDS PostgreSQL cloud database.

Once a welfare alert is generated and transmitted to the administrator cloud dashboard, it includes information on the anonymized PAGES child identifier and the Resident ID cross-reference. Using this information, the administrator can retrieve the child's identity, camp zone, tent location, and guardian contact information by querying the Resource Management database with the Resident ID. This enables immediate welfare action without storing any personally identifiable information in the PAGES database. In the following steps, a volunteer from the available volunteer list within the Resource Management database may be assigned to the respective camp to conduct a physical welfare check on the child. This architectural design ensures that any violation or exploitation of the PAGES database grants access only to anonymized identifiers and behavioral session records unrelated to any identifiable child, while maintaining the entire welfare response sequence, from automated alert generation to physical welfare check.

Encrypted Cloud Transmission

All data transfer between the Smart-Camp Box Raspberry Pi device and the AWS RDS PostgreSQL cloud database is protected by HTTPS, utilizing a minimum version of TLS 1.2. Synchronization is controlled by the wlan1 Station Mode WiFi interface only when it detects an active Internet connection and is performed only with the AWS RDS endpoint that has been authenticated. There is no plaintext data transfer over any segment of the network. The synchronization service logs in to AWS using IAM with the minimum privileges required to read and write the PAGES table, preventing unauthorized access to the Resource Management and Disaster Prediction tables even if AWS credentials are compromised.

Role-Based Access Control

PAGES has developed separate authentication schemes for each of its three interfaces. This is due to the different connectivity scenarios in which they are deployed. For the child interface, only an anonymized QR code scan from the wristband or an ID code entry is required. Children can only access the gamified task, calming activity, and cognitive content screens. Under no circumstances may children access any dashboards, administrative functions, profile information, or other child-related information via the child interface.

Local Network Isolation

The Access Point (AP) mode WiFi hotspot provided by the integrated wlan0 interface is accessible only within the 192.168.4.x subnet and provides access to the PAGES Next.js web application only. IP forwarding between the wlan0 and wlan1 interfaces is disabled at the operating system level to prevent client devices from routing traffic to the Internet via the device. All traffic within the wlan0 network is confined to the subnet and thus prevents child devices from accessing Internet-based threats and any network resource beyond the PAGES web application. The wlan1 interface in the Station mode, which provides the Internet connection for cloud synchronization, uses a different network interface and is not connected to the wlan0 interface under any circumstances.

Shared Database Access Boundaries

To ensure that data is not exposed between other components, boundaries are explicitly defined at the database level. For each component, there is a dedicated PostgreSQL role defined, and that role has access only to tables that are prefixed with that component's name. For example, the PAGES database role has only read and write access to tables that have 'pages_' as a prefix, and no access at all to tables that have 'resource_' or 'disaster_' prefixes. In cases where cross-component data access is required, i.e., in the JOIN operation between pages_alerts and resource_residents tables in the resolution of welfare alerts, data access is provided via a dedicated read-only dashboard database role that has SELECT privilege on only those columns that are necessary for alert displays, and not via any of the

roles defined for these components. This approach ensures that if an unauthorized actor gains access to the PAGES application database credentials, they cannot gain access to Resource Management identity data, and that the Resource Management component cannot gain access to PAGES behavioral session data.

Absence of Third-Party Data Sharing

No data related to child behavior or session activity is sent to any third-party service, analytics system, advertising system, or API at any point in the system's lifecycle. The PAGES application does not make any external HTTP requests from interfaces designed for child users.

4.3.2 Scalability Considerations

The PAGES architecture is designed to scale across five dimensions relevant to disaster relief deployment: concurrent users, data volume, geographic distribution, ML model complexity, and connectivity variability.

- Concurrent Users: A single Smart-Camp Box can support 15–25 concurrent child device connections. The Raspberry Pi 5 can support a Flask-based web server, SQLite for both read and write operations, and Phase 1 Z-Score Machine Learning Inference within a time bound of <3 seconds. When concurrent users exceed 25, the second Smart-Camp Box can run in parallel with the first. Both Smart-Camp Boxes can run independently, each with their own local child database.
- Data Volume: The rolling baseline architecture restricts the data storage for each child to one score update per session rather than an uncontrolled data history. SQLite can support at least 1,000 session records without impacting performance on Raspberry Pi 5. Phase 2 Isolation Forest Model Files have no significant storage requirements, as they are small, serialized Python objects (10-50 KB per child).
- Multi-Device Deployment: In this solution, each Smart-Camp Box runs independently. The cloud layer collects data from all deployed Smart-Camp Boxes. This enables regional- or national-level monitoring of multiple camps without requiring any device-to-device relationships or network meshing. There is no need to reconfigure the existing deployed devices when adding a new device.
- Scalability of the Machine Learning Model: Phase 1 of the Z-Score inference is completed within less than two seconds on-device with a negligible computational cost. Phase 2 of the Isolation Forest model training is completed in the cloud on standard CPU-based computer infrastructure. No special computational resources are needed for training. Once trained, models are deployed back to the device as a lightweight file format.
- Connectivity Independence: Our system is designed to function at full capacity regardless of internet connectivity. Hence, scalability is not affected in any way by internet connectivity. A camp in a state of complete internet disconnection can use PAGES indefinitely, thereby continuing to build up data in its local synchronization queue and retrospectively synchronize when internet connection is available, even for a brief period.

5. USER REQUIREMENTS

5.1 Users / Stakeholder Identification

Primary Users (Direct System Interaction)

Table 11: Primary Users

User	Who are they?	Role in PAGES
Grade 3-4 children	Displaced children in disaster relief camps	Complete gamified assessment, receive gate decision, engage with cognitive content
Administrators	coordinators accessing the system via the cloud	View synced session data, review anomaly alerts

Secondary Users (Indirect Interaction)

Table 12: Secondary Users

Stakeholder	Who are they?	Role in PAGES
Camp Coordinators	Senior staff managing overall camp operations	Coordinate physical intervention
Zonal Education Office	National curriculum authority	Ensure content aligns with the Grade 3-4 national framework
Smart-Camp Box System	Sibling IoT components (disaster prediction, resource management)	Provide camp context data; receive PAGES session outputs

Functional Requirements

- System shall present a gamified Go/No-Go task (tap target animals, ignore distractors) lasting 90 seconds at the start of each child session.
- System shall capture four behavioral metrics from the gamified task: accuracy rate, false tap rate, mean reaction time, and reaction time variance.
- System shall passively collect behavioral interaction signals throughout the session: idle time ratio, rapid click counts, time to first interaction, and session completion ratio.
- System shall retrieve the child's pre-computed rolling baseline score from their local profile at session start.
- System shall compute a composite psycho-attentional score (0 -100) by fusing three data sources: gamified task output (50%), passive behavioral signals (30%), and historical baseline (20%).
- System shall compare the composite score against a defined readiness threshold and issue a binary gate decision (Pass / Fail) per session per child.
- System shall deliver a calming or re-engagement support activity to any child who receives a Fail gate decision.
- System shall re-assess the child's psycho-attentional score after the support activity and issue a second gate decision.
- System shall deliver a standardized curriculum aligned with cognitive content to all children who receive a Pass gate decision.
- System shall record cognitive content performance scores and session progression data per child per session.
- System shall update the child's rolling baseline score after each session using the formula: New Baseline = (Old Baseline × 0.8) + (Current Score × 0.2).

- System shall apply Phase 1 Z-Score anomaly detection from session three onwards per child, flagging sessions where the score falls below 2.0 standard deviations from the child's personal mean.
- System shall apply Phase 2 Isolation Forest anomaly detection per child after sufficient session accumulation, replacing Phase 1 for that child.
- System shall generate a welfare alert when a child has three or more consecutive flagged sessions, or three flagged sessions within the last five sessions.
- System shall store welfare alerts locally and display them on the volunteer facilitator dashboard in plain-language, actionable format.
- System shall allow admin to mark alerts as resolved or escalate them to the camp coordinator.
- System shall store all session data, child profiles, composite scores, gate decisions, and alerts in local SQLite storage on the Smart-Camp Box device.
- System shall automatically synchronize all locally stored data to cloud storage when it detects an available internet connection
- System shall support a minimum of 15 simultaneous child device connections via the Smart-Camp Box local WiFi hotspot
- System shall maintain individual child profiles containing child ID, grade, rolling baseline score, session history, gate decision history, and alert status
- System shall allow registration of new child profiles and update basic profile information
- System shall provide the admin dashboard with their current gate status, and any pending alerts
- System shall allow remote online administrators to view synced session data, child progress, and alert logs via the cloud dashboard

5.3 Non-Functional Requirements

Performance

- System shall deliver gate decisions within 5 seconds of gamified task completion
- System shall maintain response times under 3 seconds for all child-facing screens under 15 concurrent users
- System shall complete local SQLite read/write operations within 1 second per session record
- System shall complete cloud synchronization of all pending session records within 60 seconds of internet connectivity detection
- Z-Score anomaly calculation shall be completed within 2 seconds per session

Security

- All child profile data shall be stored using anonymized child ID codes; no real names shall be stored on the device
- Cloud-synced data shall be transmitted over HTTPS with TLS 1.2 or higher encryption
- Remote administrator cloud access shall require role-based authentication
- No child behavioral data shall be shared with third parties outside the Smart-Camp Box research framework

Usability

- Child-facing gamified task interface shall use icon-based instructions and simple reading required

- All child-facing screens shall use a minimum font size of 18pt with high-contrast colors suitable for shared tablet use in outdoor/bright-light conditions
- Admin dashboard welfare alerts shall be written in plain English with no technical or ML terminology visible
- An admin with no prior technical training shall be able to navigate the dashboard and action an alert within 5 minutes of first use

Reliability

- System shall operate fully offline or with low network connectivity.
- System shall recover all in-progress session data following an unplanned power interruption without corruption
- System shall achieve a minimum 99% uptime during active camp operational hours

Scalability

- System shall support a minimum of 100 individual child profiles stored per device
- Local SQLite database shall handle a minimum of 1,000 session records without performance degradation

Compatibility

- Child-facing web applications shall function on any device with a modern browser (Chrome, Firefox, Safari)
- System shall operate on Raspberry Pi 5 as the minimum hardware specification
- System shall support child device screen sizes from 5 inches (smartphone) to 10 inches (tablet)

7. COMMERCIALIZATION PLAN

7.1 Target market and customer persona

The Smart-Camp Box framework taps into an identified and highly significant market space characterized by the convergence of humanitarian technology, disaster management, and educational continuity.

7.1.1 Primary Market - Government Disaster Management Authorities

The primary market is the disaster management authority responsible for activating, managing, and provisioning disaster relief camps at the national and divisional levels. In the case of Sri Lanka, the Disaster Management Centre under the Ministry of Public Security is the primary market, with the Divisional Secretariat of Kolonnawa specifically engaged as a direct stakeholder for this research. They require a deployable solution that is offline-enabled and capable of concurrently addressing the identification of displaced populations, distribution of resources, welfare assessment, and site-level disaster risk assessment without the use of telecommunications infrastructure.

7.1.2 Secondary Market - Zonal and Provincial Education Authorities

The secondary market is the educational authority that is responsible for ensuring the continuity of the curriculum of the displaced population. In the case of Sri Lanka, the Zonal Education Office is the secondary market, with the ZEO Homagama specifically engaged as a direct stakeholder for the PAGES component. They require a solution that can deliver certified educational content to Grade 3 and Grade

4 students without the use of the Internet, while concurrently assessing the psychological welfare of the child and reporting to the appropriate adult authority if the child is in need.

7.1.3 Tertiary Market - International Humanitarian Organizations

The tertiary market is the international humanitarian authority that is responsible for conducting educational and welfare programs in disaster-affected regions. UNICEF, Save the Children, UNHCR, and similar international entities operate educational continuity programs in disaster-affected regions across the globe.

7.1.4 Customer Personas

Table 13: Customer Personas

Persona	**Role**	**Primary Need from Smart-Camp Box**
Camp Coordinator	Divisional Secretariat officer managing a single relief camp	Real-time resource allocation, population headcount, welfare alerts, and site flood risk
Zonal Education Officer	The ZEO official is responsible for the displaced children's education	Curriculum-verified educational delivery, child welfare monitoring, and attendance data
Camp Volunteer or Teacher	On-ground facilitator with no technical background	Simple device operation, QR scan enrolment, plain-language welfare alerts
UNICEF Field Officer	Humanitarian agency program officer	System compliance with INEE standards, data protection, deployment evidence

7.2 Value Proposition

7.2.1 Smart-Camp Box - Integrated Value Proposition

Smart-Camp Box is the only portable, offline-first IoT system that simultaneously manages displaced population identification, resource distribution, educational continuity, child welfare monitoring, and site-specific disaster risk prediction, all from a single Raspberry Pi 5 device deployable within minutes of camp activation, with zero internet dependency for core operations.

7.2.2 Per-Component Value Propositions

Table 14: Per-Component Value Propositions

Component	**Problem Solved**	**Unique Value Delivered**
Resource Management	Unverified identity leads to duplicate aid distribution and resource waste	Pre-disaster fingerprint registration and wristband QR barcode system ensures every resource distribution is linked to a verified, unique individual, preventing duplication without internet

Component	Problem Solved	Unique Value Delivered
PAGES	Traumatized children are pushed into academic content they cannot absorb, wasting educational sessions and potentially worsening distress	Checks psycho-attentional readiness before every session, gates content access, delivers calming support when needed, and alerts welfare authorities when a child shows persistent distress patterns, the first system to do this
Smart Disaster Monitoring	Regional flood warnings do not tell camp managers whether their specific camp will be flooded	GPS-anchored, altitude-aware flood and landslide probability model gives camp-level

7.2.3 Combined System Value

There are no competing products that provide all three functionalities in a single product that can be used offline. The bundling of identity management, educational welfare monitoring, and site risk prediction in a single Smart-Camp Box provides a system whose total value is greater than its parts. This is because a coordinator can simultaneously check attendance at a camp, check that children are receiving proper educational welfare, and check if the site is risky.

7.3 Revenue Model

Considering the humanitarian and public sector nature of the target customer segment, the Smart Camp Box model will be based on a public procurement and licensing model rather than a consumer subscription model.

Revenue Stream 1 - Hardware Sales to the Government

The Smart Camp Box hardware will be sold to the Disaster Management Centre or an equivalent government ministry as a single hardware unit. The hardware will include a Raspberry Pi 5, sensors, WiFi hardware, battery, solar panel, wristband printer, and pre-installed software. These hardware units will be sold to the government through a tender process and will be deployed to the Divisional Secretariats for storage in pre-positioned stockpiles for disaster preparedness.

Revenue Stream 2 - Software Licensing to the Educational Sector

The PAGES educational software will be sold to the Zonal Education Offices as a separate licensing model for offline educational continuity, not requiring the Smart Camp Box context. The Zonal Education Offices will be able to use PAGES to maintain continuity in educational activities in schools affected by non-disaster-related emergencies such as floods, dengue fever outbreaks, and other emergency school closures.

Revenue Stream 3 - Annual Maintenance Contract

The annual maintenance contract will include software upgrades, curriculum content upgrades for PAGES with changes to the NIE curriculum, hardware upgrades, and remote tech support for the government deployers.

Revenue Stream 4 - Licensing to Humanitarian Organizations

International humanitarian organizations, such as UNICEF and Save the Children, could license the complete Smart-Camp Box software stack for deployment in their hardware procurement programs in

other countries. This revenue stream is based on a royalty per country deployment, thus facilitating international revenue without the need for the research team to implement hardware on a global scale.

Revenue Stream 5 - Research and Consultancy

The validated system, research findings, and deployment data inform consultancy services to government disaster preparedness programs, regional humanitarian technology programs, and international development organizations interested in technology design in displacement settings.

7.4 Cost estimation

7.4.1 Per-Unit Hardware Cost (Single Smart-Camp Box)

Table 15: Per-Unit Hardware Cost

Item	**LKR**
Raspberry Pi 5 (8GB)	24,760
MicroSD card (64GB)	4,640
Rainfall sensor	3,710
Ground movement sensor	5,570
GPS module	6,810
Fingerprint scanner	10,830
Wristband QR printer	13,930
Wristband stock (500 units)	6,190
Dual-band WiFi adapter	5,570
LiFePO4 battery pack (12V 20Ah)	17,020
Solar panel (20W portable)	12,380
Weatherproof enclosure + mounting	9,280
Tablets × 5 (child devices)	54,160
Cables, connectors, misc.	6,190

Total Hardware Per Unit: LKR 181,070

7.4.2 Software Development Cost (One-Time)

Table 16: Software Development Cost

Item	**LKR**
PAGES component (6 months, 1 dev)	1,857,000
Resource Management component	1,857,000
Disaster Monitoring component	1,857,000
Integration, testing, deployment tooling	928,500
AWS cloud infrastructure setup	154,750

Total Software Development: ~ LKR 6,654,250

7.4.3 Annual Operating Cost (Per Deployment)

Table 17: Annual Operating Cost

Item	**LKR/year**
AWS RDS PostgreSQL	74,280
AWS Lambda + S3	18,570
AWS Amplify (dashboard hosting)	37,140
Software maintenance + updates	371,400
Curriculum content updates	92,850

Total Annual Operating Cost: LKR 594,240

7.5 Pricing strategy

7.5.1 Government Hardware Supply

- Price per Smart-Camp Box (to government): **LKR 371,400**
- Markup**:** ~2× over component cost (covers assembly, testing, logistics, deployment support)
- National deployment (300 camps): **LKR 111,420,000**

7.5.2 PAGES Educational Licence

- Annual licence per ZEO: **LKR 247,600**
- Sri Lanka (97 ZEOs): **LKR 23,990,200** per year

7.5.3 Annual Maintenance Contract

- Per Smart-Camp Box unit: **LKR 46,425** per year

- National deployment (300 units): **LKR 13,927,500** per year

7.5.4 International Humanitarian Licensing

- Country-level licence (per year): **LKR 1,547,500**
- **Covers:** Right to deploy Smart-Camp Box software stack on NGO's own hardware procurement

7.6 Competitive advantage

7.6.1 Stakeholder Integration Advantage

The Smart-Camp Box system was designed in close collaboration with the Divisional Secretariat of Kolonnawa, as the government disaster management stakeholder, and the Zonal Education Office of Homagama, as the educational authority stakeholder. This close working with stakeholders gives the Smart-Camp Box system three competitive advantages that are difficult for new entrants to achieve: (1) well-validated needs based on real-world operational experience, (2) letters of support that can be used to strengthen government procurement bids, and (3) an existing method of validation through existing institutional connections.

7.6.2 Research Validation Advantage

The PAGES component of the Smart-Camp Box is the only educational welfare system for post-disaster children, with published validation research for its psycho-attentional assessment layer, its gating algorithm, and its machine-learning anomaly-detection pipeline. This published research base provides a defensible technological platform for government procurement decisions, where evidence of the system's effectiveness is required before deployment by government or humanitarian agencies.

7.6.3 Deployment Simplicity Advantage

The complete Smart-Camp Box system is designed to be deployable and operable by a non-technical volunteer or government officer with no information technology skills whatsoever. The QR wristband scan, the web-based child interface, the plain-language welfare notifications, and the automated cloud syncing require no command-line access, no servers to manage, and no IT skills. Therefore, the adoption threshold is significantly lower than for enterprise-level IoT solutions, which typically require specialist deployment teams.

7.7 Intellectual property considerations

7.7.1 Software and Algorithm Intellectual Property

The three technical innovations identified as part of the Smart-Camp Box system constitute the primary intellectual properties. The Resource Management Component's Semantic-Aware Adaptive MQTT Message Prioritization framework is a novel communication protocol architecture with no known antecedent or analogue, making it the strongest candidate for formal patent protection. The PAGES component's psycho-attentional composite scoring formula, gating function, and two-stage child welfare detection via the Isolation Forest algorithm constitute novel machine-learning-based techniques that may be formally protected as software copyright. However, the combination of algorithms may also be formally protected as a patent. The Disaster Monitoring component's GPS-altitude-anchored camp-specific flood prediction is a novel prediction methodology that may be formally protected as software copyright or as a patent.

7.7.2 Recommended IP Actions

Before submitting the research paper to the public domain, the group should file provisional patent applications with the National Intellectual Property Office of Sri Lanka under the Intellectual Property Act No. 36 of 2003. The group should file provisional patent applications for the three technical innovations. Provisional patents are cost-effective, establish the priority date, and allow the group 12 months to file or evaluate the commercial viability of the invention before formal filing.

The Smart-Camp Box software codebase should be formally registered as original literary work under the copyright provisions of the same Act. Automatic protection is granted upon creation of the software, and formal registration is advisable to establish the evidentiary value should an infringement issue arise.

7.7.3 University Intellectual Property Policy

The system was developed as part of a university-supervised final-year project; as a result, the intellectual property policy of the university to which the project was submitted applies. In the case of intellectual property developed by a student as part of supervised academic research, ownership would typically vest jointly with the university or solely in the university, and the student would hold a license. The research team would have to consider the intellectual property policy of the university and negotiate a commercialization agreement with the university's technology transfer office before entering into any license agreements, to ensure the ownership, revenue split, and the rights of the student team to operate a spinoff venture using the system developed by the student team.

7.7.4 Open-Source Strategy

Another alternative to the patent-based commercialization model is to adopt a dual-license open-source model, whereby the Smart Camp Box software would be licensed as open source under a copyleft license, such as the GPL v3, for humanitarian and non-commercial use, and a commercial license would be granted to government and private-sector deployers. This would maximize humanitarian impact by enabling NGOs and foreign governments to use the system and generate revenue from its sale to the government of Sri Lanka. The PAGES curriculum content layer would remain proprietary and licensed separately to ensure compliance with the ZEO content quality standard.

7.7.5 Stakeholder Acknowledgement in IP

The Divisional Secretariat of Kolonnawa and the Zonal Education Office of Homagama provided access, operational requirements, and validation support for the system, and these stakeholders would have a significant influence on the system developed. These stakeholders would have to be acknowledged for their contribution to the system, and the system would have to be considered for the commercialization structure, particularly for the system being deployed within the jurisdiction of the operations of these stakeholders.

8. BUDGET AND JUSTIFICATION

8.1 Hardware Costs

Table 18: Hardware costs

Item	Specification	Total	Justification
Raspberry Pi 5	4GB RAM	Rs. 36500	Primary edge computing device for Smart-Camp Box. 4GB RAM is required for Flask server, SQLite operations, and ML inference simultaneously. Two units: one for development and testing, one for pilot deployment.
MicroSD Card	64GB Class 10 / U3	Rs. 2100	Local operating system, application files, SQLite database, and ML model file storage. 64GB provides sufficient headroom for 1,000+ session records and model files.
USB WiFi Adapter	TP-Link TL-WN722N (wlan1 - STA mode)	Rs. 6000	Dedicated WiFi adapter for Station mode internet detection and cloud synchronization. Required as the Raspberry Pi 5 built-in WiFi (wlan0) is reserved for the Access Point hotspot.
Solar Charging Panel	20W, 5V USB output	Rs. 6000	Sustainable field power source for battery recharging in camp environments without grid access. Supports indefinite deployment duration.
Hardware Sub Total:		**Rs. 50600**	

8.2 Software Licenses

Table 19: Software costs

Item	Type	Cost	Justification
Raspberry Pi OS	Open source (free)	Rs. 0	Official Raspberry Pi Foundation operating system. Fully supported, stable, and optimised for Pi 4 hardware. No license cost.
Python 3.10+	Open source (free)	Rs. 0	Primary backend language for Flask server, SQLite operations, Z-Score calculation, and Isolation Forest ML inference.
Flask Web Framework	Open source (free)	Rs. 0	Lightweight Python web framework for serving the PAGES web application and REST API. Suitable for embedded IoT deployment on Raspberry Pi hardware.

Item	Type	Cost	Justification
SQLite	Open source (free)	Rs. 0	Embedded relational database requiring no separate server process. Ideal for offline IoT deployment with low overhead and zero configuration.
scikit-learn (Python)	Open source (free)	Rs. 0	Provides Isolation Forest implementation for Phase 2 anomaly detection. Mature, well-documented, and computationally lightweight for RPi deployment.
NumPy / Pandas	Open source (free)	Rs. 0	Numerical computation libraries for Z-Score calculation, feature vector assembly, and data preprocessing.
Visual Studio Code	Free (Microsoft)	Rs. 0	Primary development IDE for Python, HTML, CSS, and JavaScript development across all PAGES components.
Git / GitHub	Free tier	Rs. 0	Version control and code repository management. GitHub's free tier is sufficient for a single-developer research project.
Figma	Free tier	Rs. 0	UI mockup and prototyping tool for child-facing interface design and volunteer dashboard wireframes. The free tier is sufficient for the research prototype scope.
Software Sub Total		**Rs. 0**	

8.3 Cloud Services

Table 20: Cloud services costs

Service	Provider	Plan	Monthly Cost	Duration	Total	Justification
Cloud Database & Storage	Supabase (PostgreSQL)	Free tier	Rs. 0	8 months	Rs. 0	Estimated 30–50 children × 30 sessions = ~1,500 records.
Cloud Functions / ML Training	Google Colab	Free tier	Rs. 0	8 months	Rs. 0	Models are small (10–50 KB each), and training time is under 30 seconds per child.
Admin Dashboard Hosting	Vercel	Free tier	Rs. 0	8 months	Rs. 0	Remote administrator dashboard hosted as a static web application on Vercel free tier. r

Service	Provider	Plan	Monthly Cost	Duration	Total	Justification
Domain Name	Namecheap	.com / 1 year	Rs. 2113	12 months	Rs. 2113	Custom domain for research project website submission
SSL Certificate	Let's Encrypt	Free	$0	8 months	Rs. 0	Required for TLS 1.2+ encrypted transmission between Smart-Camp Box and cloud infrastructure.
Backup Cloud Storage	Google Drive	Free (15GB)	$0	8 months	Rs. 0	Research data backup, document version control, and supervisor sharing.
Cloud Sub Total					**Rs. 2113**	All required cloud services operate within free

8.4 Human Effort Cost

Table 21: Human Effort Cost

Role	Activity	Hours	Rate	Total
Researcher (Self)	Proposal writing, system design, development, testing, thesis writing	896	-	-
Supervisor	Academic supervision, feedback sessions, and review meetings	30	-	-
Co-supervisor	Academic supervision, feedback sessions, and review meetings	30	-	-
Divisional Secretary	Share pain points and requirements	3	-	-
Primary Teacher	Educational requirements and share students' behaviors	2	-	-
Human Effort Sub Total				Rs. 00

8.5 Miscellaneous and Contingency

Table 22: Miscellaneous and Contingency Costs

Item	Cost (LKR)	Justification
Travel to meet stakeholders	Rs. 2700	Discussion to align with the design project.
Meet the supervisor and co-supervisor	Rs. 8000	Get academic support and guidance.
Printing Proposal, thesis drafts, consent forms	Rs. 400	Physical document printing for supervisor review, ethical clearance submission, and pilot study consent documentation
Contingency (10% of hardware)	Rs. 50000	Hardware replacement buffer for component failure during field deployment. The Raspberry Pi and USB adapter are most likely to require replacement.
Miscellaneous Sub Total	**Rs. 61100**	

Total Budget = 50600+2113 +61100

= 113813

8. WORK BREAKDOWN STRUCTURE (WBS)

9.1 Task Breakdown Per Member

Member 1 - Hansana G P | IT22064868 Sub-Component: Smart Environmental Risk Monitoring and Location-Aware Flood and Landslide Prediction

- GPS/GNSS altitude and location detection module
- Multi-sensor data collection (rainfall, soil moisture, humidity, ground movement)
- Watershed and water path geospatial analysis using QGIS and OpenStreetMap
- External data integration - Irrigation Department River levels and OpenWeatherMap API
- Weighted multi-factor disaster risk prediction model development
- Centralized multi-camp administrative dashboard (Next.js, Node.js, AWS IoT Core, PostgreSQL)
- Field prototype assembly, sensor calibration, and stakeholder validation

Member 2 - Kashmira S S | IT22059772 Sub-Component: Human and Resource Demand Prediction

- Intelligent crowd detection and human counting using camera-based IoT and ML
- Demographic estimation from the detected population for food, water, and medicine demand prediction
- Edge-cloud hybrid model for real-time and future resource forecasting
- Integration of human detection modules with a cloud-based dashboard
- Comparison of AI-predicted demand against conventional manual estimation practices

Member 3 - Deerasinghe A D S N S | IT22065780 Sub-Component: Psycho- Attentional Gated Education System

- Design and implement the gamified Go/No-Go task (Tap the Animal - 90 seconds)
- Calculate Cronbach's alpha across baseline sessions for test-retest reliability
- Implement binary gate decision logic with a threshold at 55/100
- Implement calming breathing activity delivered on the first fail decision
- Implement re-assessment flow after calming activity
- Implemented the second gate decision and confirmed the fail logic
- Implement consecutive_low_scores counter in SQLite sessions table
- Conduct chi-square analysis comparing pass rates across session conditions
- Calculate session recovery rate from gate decision logs
- Design Grade 3 and Grade 4 literacy and numeracy activities aligned to NIE curriculum
- Submit content package to ZEO Homagama designated reviewer for alignment verification
- Implement cognitive content delivery interface with session completion tracking
- Calculate curriculum alignment rating from ZEO reviewer checklist
- Run linear regression on cognitive content scores across sessions per child
- Implement Phase 1 Z-Score computation per child from session 3 onwards
- Implement alert trigger logic - 3 consecutive flagged or 3 of the last 5 flagged
- Implement Phase 2 Isolation Forest training pipeline on AWS Lambda
- Deploy Next.js frontend and Node.js Express backend on Raspberry Pi 5
- Configure SQLite with WAL mode enabled for data integrity on power interruption
- Implement sync queue table and sync service in Node.js
- Implement connectivity watcher with automatic sync trigger on internet detection
- Run 30 simulated offline sessions with network interface disabled
- Run Apache JMeter load test simulating 15 concurrent child device connections
- Conduct force-shutdown mid-write test and verify SQLite integrity after restart
- Verify 100% record count match between SQLite and PostgreSQL after sync

9.2 Timeline Alignment

Table 23: Timeline Alignment

Activity	**Semester**	**Weeks**
Topic selection, literature review, proposal writing	Semester 1	W1 - W10
System architecture and component design (all members) Semester 1 W10 – W18	Semester 1	W10-W18
Hardware procurement and Smart Disaster Box	Semester	W14
Individual sub-component development (parallel)	Semester 1-2	W18 - W32
Sub-component integration and system-level testing	Semester 2	W30 - W38
Stakeholder validation, final testing, and refinement	Semester 2	W36 - W42
Final report, research paper, viva preparation	Semester 2	W38 - W48

9.3 Balanced Workload Distribution

- Deerasinghe A D S N S- Anomaly detection, welfare alert, Smart Camp Box hardware, documentation
- Hansana G.P- Environmental risk prediction, GPS altitude model, multi-camp dashboard, Smart Disaster Box hardware, cloud platform setup
- Kashmira S S- Human detection, crowd counting, resource demand forecasting Camera integration, cloud dashboard data feeds

All three members share responsibility for the following cross-cutting tasks:

Joint system architecture design

Smart Disaster Box hardware integration

System-level integration testing in

Co-authorship of the final research paper and viva presentation.

9. GANTT CHART

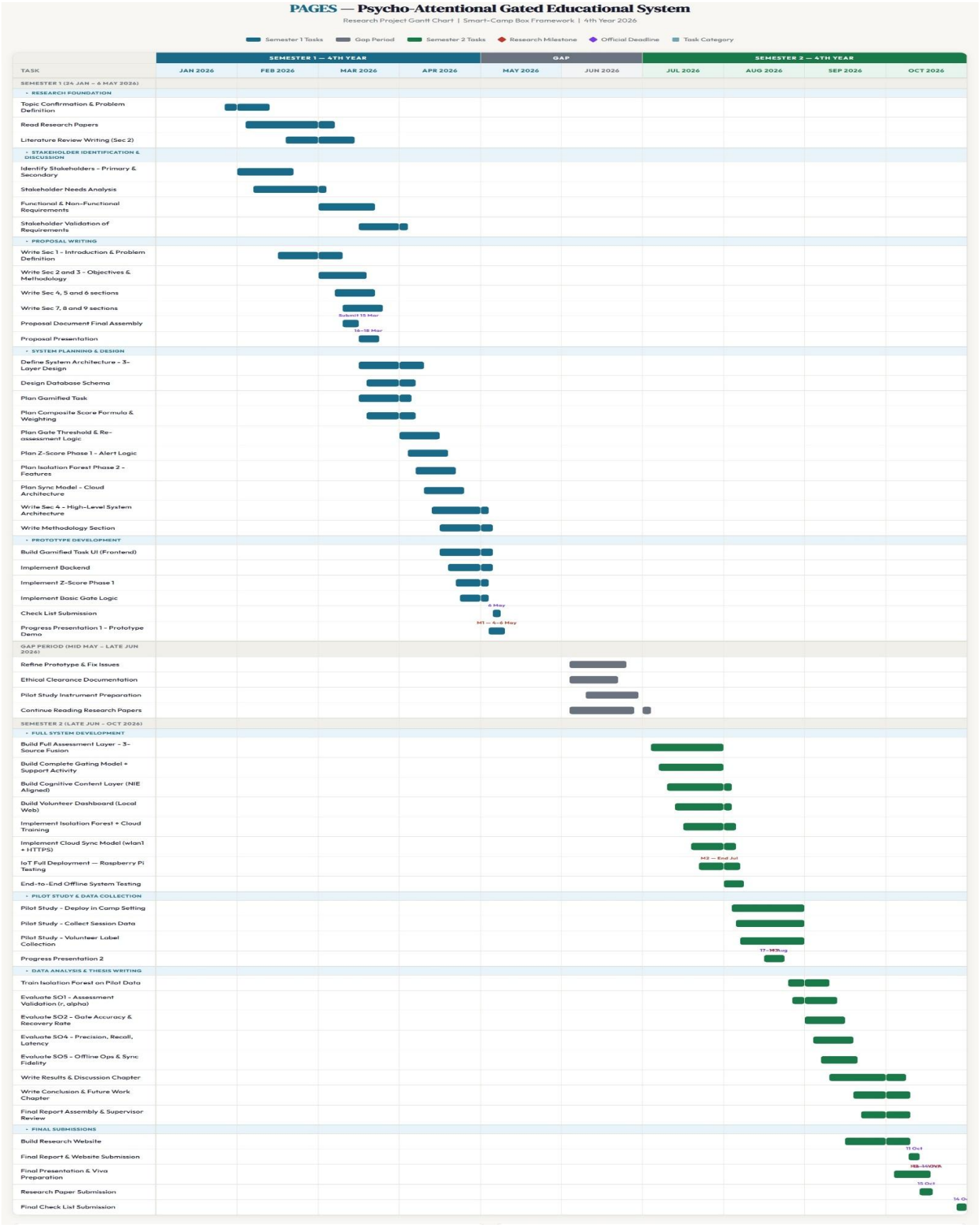


Figure 8: Gantt chart

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11. APPENDICES

Appendix A - Stakeholder Survey Instruments

A.1 Volunteer Facilitator Readiness Rating Form

Used during the pilot study to collect ground-truth child readiness labels for design and to validate a Combined Psycho-Attentional Assessment Layer and implement and evaluate the Psycho-Attentional Gating Model validation.

Instructions to Volunteer: At the beginning of each child's session, observe the child for approximately 60 seconds before they begin the gamified task. Then circle the rating that best describes how the child currently appears.

Section 1 - Psychological State

How does the child appear emotionally at this time?

Section 2 - Attentional State

How focused does the child appear right now?

Section 3 - Overall Readiness

In your judgment, is this child ready to start a learning activity right now?

- Definitely ready
- Probably ready
- Unsure
- Probably not ready
- Definitely not ready

A.2 Welfare Alert Usability Questionnaire

Completed by Admin after receiving their first welfare alert from PAGES.
Child ID Code: C0001

1. When you saw the alert, did you understand what it was telling you?
2. Did you understand what action you should take after seeing the alert?
3. How would you rate the clarity of the alert message? (1 = very unclear, 5 = very clear)
4. How would you rate the usefulness of the suggested action? (1 = not useful, 5 = very useful)
5. Did you physically go and check on the child after receiving the alert?
6. After checking on the child, was the alert accurate?
7. Any comments or suggestions about the alert system:

A.3 Child Cognitive Content Experience Observation Form

Completed by a volunteer observer during cognitive content sessions.

Child ID Code: C0001 Date: 07/07/26 Session No.: 3

Observation Rating 1-5

- Child appeared to understand the activity instructions
- Child engaged with the activity independently
- Child completed the activity without distress
- An activity appeared appropriate for the child's age

Appendix B - Detailed System Architecture Diagrams

B.1 PAGES Three-Layer Architecture Diagram - See Figure 5.

B.2 Session Pipeline Flowchart

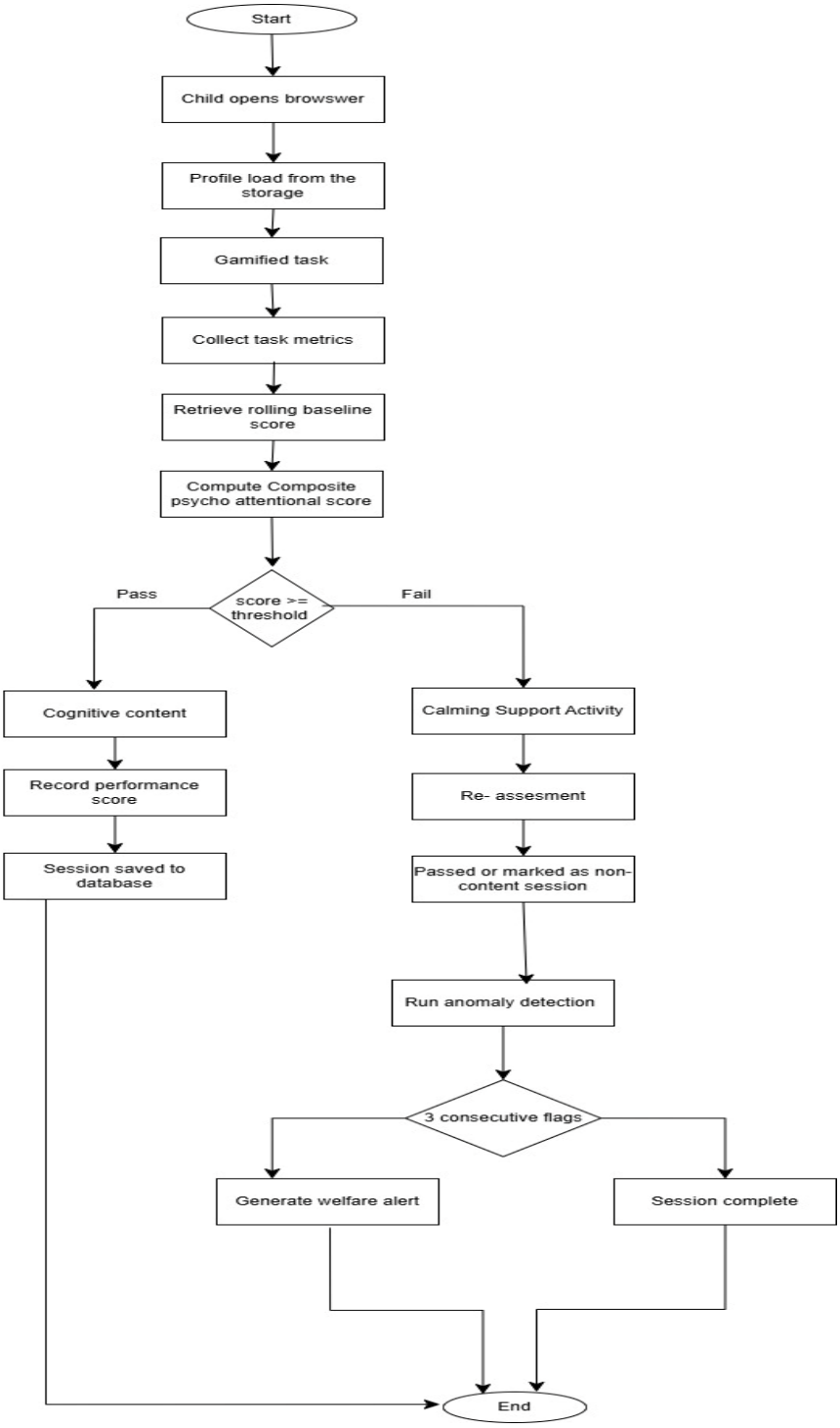


Figure 9: Session flowchart

Appendix C - Preliminary UI Mockups

Full interactive mockups available in the Figma project file submitted alongside this document.

C.1 Child-Facing Gamified Task Screen

Icon-only instructions - no reading required

Large touch targets (min 80×80px)

High contrast background Progress bar only - no score shown to child

C.2 Gate FAIL - Support Activity Screen

Calm color palette (soft blue/green)

No timers visible

Positive reinforcement only

Simple animated breathing guide

C.3 Admin Dashboard - Alert View

Plain language - no ML terminology

Child ID only - no real name

Clear action steps, Two-button response

Appendix D - Risk Analysis Register

Table 24: Risk Analysis Register

Risk ID	Risk Description	Likelihood	Impact	Severity	Mitigation Strategy	Contingency
R01	Raspberry Pi hardware failure during pilot deployment	Medium	High	High	Two units purchased - one primary, one backup. Both were configured identically before deployment.	Switch to the backup device immediately. Database data is preserved on a microSD card.
R02	Insufficient pilot study participants - fewer than 30 children with 5+ sessions each	Medium	High	High	Begin stakeholder access negotiation early in the Semester 1 gap period. Identify multiple camp sites.	Supplement with simulated session data for model training
R04	Gamified task found to be too complex or frightening for Grade 3 children	Low	High	Medium	Conduct informal usability testing with 3–5 children before the formal pilot. Iterate on animal selection and instructions.	Simplify to a single-animal tap task. Reduce session duration from

						90 to 60 seconds.
R05	Composite score shows low correlation with volunteer ratings ($r < 0.65$)	Medium	High	High	Adjust feature weights iteratively based on early pilot data. Consult D'Mello and Graesser's weighting literature.	Lower target threshold. Document partial validation. Reframe as an exploratory study.
R06	Isolation Forest precision below 0.75 on pilot data	Medium	Medium	Medium	Tune the contamination parameter on the training data. Review feature selection. Add session_streak_low as an additional feature.	Fall back to Z-Score only for anomaly detection in the final evaluation. Document Phase 2 as future work.
R07	Raspberry Pi WiFi hotspot instability under 15+ concurrent connections	Low	Medium	Medium	Test with 20 simulated concurrent connections in the development environment before the pilot. Enable QoS on the hotspot.	Reduce pilot to 10 concurrent users. Schedule sessions in smaller groups.
R08	Cloud sync failure - data loss between device and cloud	Low	High	Medium	Sync queue never deletes records until cloud acknowledgment is received. Retry logic with exponential backoff. Manual export function for emergency backup.	Manual microSD card data extraction as an emergency backup for pilot session records.

Appendix E - Ethical Clearance Documents

E.1 Research Ethics Checklist

Table 25: Ethical considerations

Ethical Consideration	**Status**	**Details**
Involves human participants	Yes	Grade 3-4 children (ages 7-9)
Participants are minors	Yes	All child participants under 18 - parental/guardian consent required
Vulnerable population	Yes	Disaster-affected, displaced children - enhanced protections applied

Ethical Consideration	Status	Details
Sensitive data collected	Yes	Behavioral and psychological state data
Data anonymization	Yes	Child ID codes only - no names in system
Informed consent required	Yes	From parent/guardian AND from child (assent form)
Right to withdraw	Yes	The child may stop at any time without consequence
Risk of psychological harm	Low	Gamified tasks are non-stressful; calming activities reduce rather than induce distress
Ethical clearance required	Yes	University Research Ethics Committee submission required before pilot study

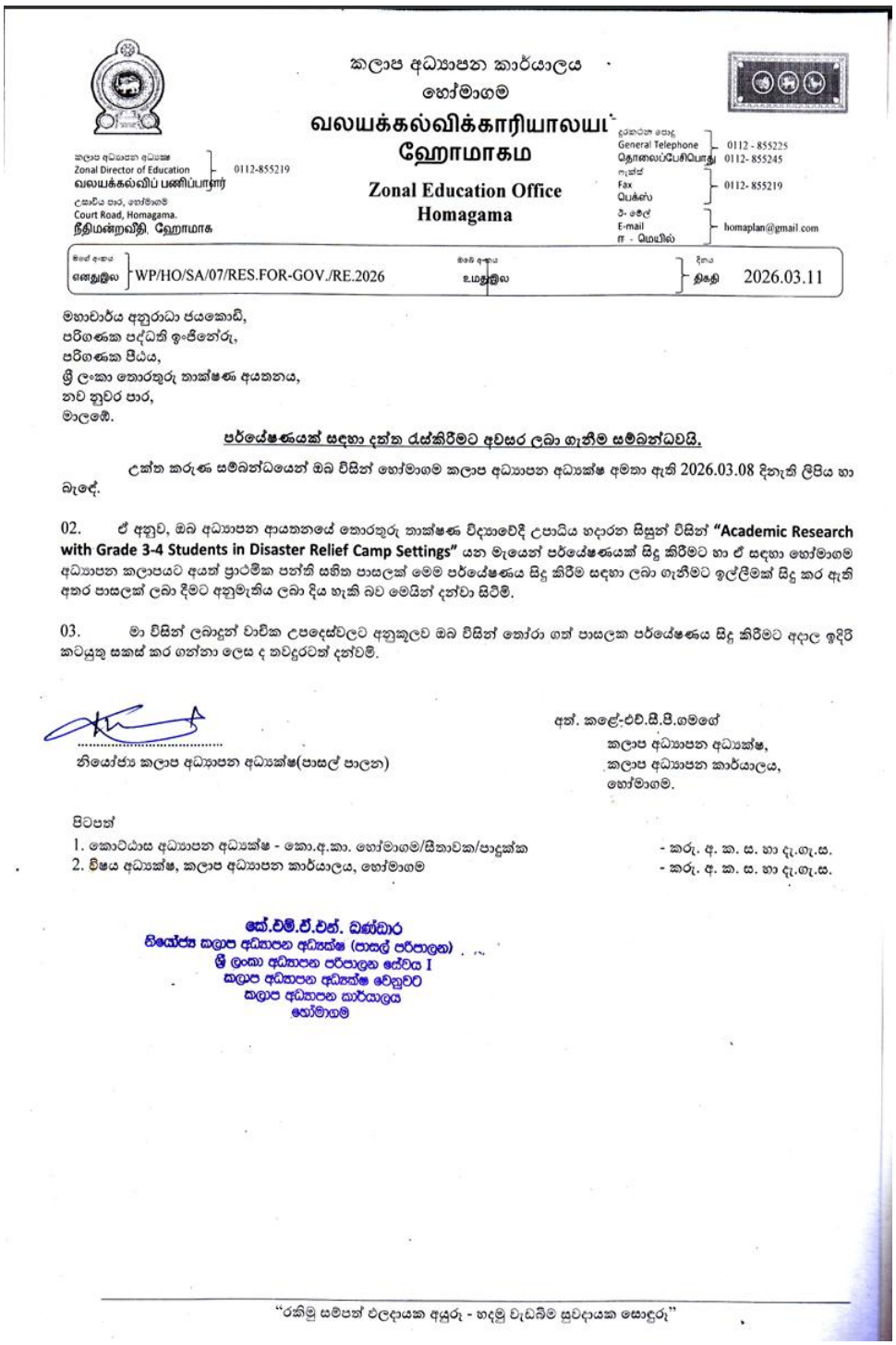
E.2 Informed Consent Form - Parent / Guardian

- What this research is about
- What will your child do
- What data is collected
- How data is protected
- Risks

E.3 Child Assent Form (Age-Appropriate Language)

- What you will do
- You can stop at any time
- We will not use your name

E.4 Permission from ZEO Homagama



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வலயக்கல்விக்காரியாலயம்
ஹோமாகம

Zonal Education Office
Homagama

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Zonal Director of Education
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எனதுஇல
WP/HO/SA/07/RES.FOR-GOV./RE.2026

ඔබේ අංකය
உமதுஇல

දිනය
திகதி
2026.03.11

මහාචාර්ය අනුරාධා ජයකොඩි,
පරිගණක පද්ධති ඉංජිනේරු,
පරිගණක පීඨය,
ශ්‍රී ලංකා තොරතුරු තාක්ෂණ අයතනය,
නව නුවර පාර,
මාලඹේ.

පර්යේෂණයක් සඳහා දත්ත රැස්කිරීමට අවසර ලබා ගැනීම සම්බන්ධයි.

උක්ත කරුණ සම්බන්ධයෙන් ඔබ විසින් හෝමාගම කලාප අධ්‍යාපන අධ්‍යක්ෂ අමතා ඇති 2026.03.08 දිනැති ලිපිය හා බැඳේ.

02. ඒ අනුව, ඔබ අධ්‍යාපන ආයතනයේ තොරතුරු තාක්ෂණ විද්‍යාවේදී උපාධිය හදාරන සිසුන් විසින් **"Academic Research with Grade 3-4 Students in Disaster Relief Camp Settings"** යන මැයෙන් පර්යේෂණයක් සිදු කිරීමට හා ඒ සඳහා හෝමාගම අධ්‍යාපන කලාපයට අයත් ප්‍රාථමික පන්ති සහිත පාසලක් මෙම පර්යේෂණය සිදු කිරීම සඳහා ලබා ගැනීමට ඉල්ලීමක් සිදු කර ඇති අතර පාසලක් ලබා දීමට අනුමැතිය ලබා දිය හැකි බව මෙයින් දන්වා සිටිමි.

03. මා විසින් ලබාදුන් වාචික උපදෙස්වලට අනුකූලව ඔබ විසින් තෝරා ගත් පාසලක පර්යේෂණය සිදු කිරීමට අදාල ඉදිරි කටයුතු සකස් කර ගන්නා ලෙස ද තවදුරටත් දන්වමි.

.....
නියෝජ්‍ය කලාප අධ්‍යාපන අධ්‍යක්ෂ(පාසල් පාලන)

අත්. කළේ-එච්.සී.සී.ගමගේ
කලාප අධ්‍යාපන අධ්‍යක්ෂ,
කලාප අධ්‍යාපන කාර්යාලය,
හෝමාගම.

පිටපත්
1. කොට්ඨාස අධ්‍යාපන අධ්‍යක්ෂ - කොට්ඨාස අධ්‍යාපන කාර්යාලය, හෝමාගම/සීතාවක/පාදුක්ක - කරු. අ. ක. ස. හා දැ.ගැ.ස.
2. විෂය අධ්‍යක්ෂ, කලාප අධ්‍යාපන කාර්යාලය, හෝමාගම - කරු. අ. ක. ස. හා දැ.ගැ.ස.

කේ.එම්.ඒ.එන්. ඛණ්ඩාර
නියෝජ්‍ය කලාප අධ්‍යාපන අධ්‍යක්ෂ (පාසල් පරිපාලන)
ශ්‍රී ලංකා අධ්‍යාපන පරිපාලන සේවය I
කලාප අධ්‍යාපන අධ්‍යක්ෂ වෙනුවට
කලාප අධ්‍යාපන කාර්යාලය
හෝමාගම

"රකිමු සම්පත් ඵලදායක අයුරු - හදමු වැඩබිම සුවදායක සොඳුරු"

Figure 10: ZEO Permission Letter